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**Theme**

**Leveraging Deep Learning Techniques for Advanced  
Recommender Systems**

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## Dedications

*To my parents,  
for every sacrifice they made so I could reach this day.*

*To my family and friends,  
for their constant support and encouragement.*

**Farouk Ouled Meriem**

# Contents

|                                                                               |           |
|-------------------------------------------------------------------------------|-----------|
| <b>Acknowledgements</b>                                                       | <b>i</b>  |
| <b>Dedications</b>                                                            | <b>ii</b> |
| <b>General Introduction</b>                                                   | <b>1</b>  |
| <b>1 Foundations and State of the Art</b>                                     | <b>4</b>  |
| 1.1 Introduction to Recommendation Systems . . . . .                          | 4         |
| 1.2 Sequential Recommendation . . . . .                                       | 5         |
| 1.3 Sequence-Based Models . . . . .                                           | 6         |
| 1.3.1 GRU4Rec . . . . .                                                       | 7         |
| 1.3.2 Caser . . . . .                                                         | 7         |
| 1.3.3 SASRec . . . . .                                                        | 7         |
| 1.3.4 BERT4Rec . . . . .                                                      | 7         |
| 1.4 Graph-Based Models / GNN . . . . .                                        | 8         |
| 1.4.1 LightGCN . . . . .                                                      | 9         |
| 1.5 LLMs Applied to Sequential Recommendation . . . . .                       | 10        |
| 1.6 Evaluation Metrics . . . . .                                              | 11        |
| 1.6.1 Hit Ratio at $K$ . . . . .                                              | 11        |
| 1.6.2 Normalized Discounted Cumulative Gain at $K$ . . . . .                  | 12        |
| 1.6.3 Mean Reciprocal Rank . . . . .                                          | 12        |
| 1.7 Conclusion . . . . .                                                      | 13        |
| 1.8 Identified Gaps and Positioning of Contributions . . . . .                | 13        |
| <b>2 Length-Adaptive Hybrid GNN-Transformer for Sequential Recommendation</b> | <b>14</b> |
| 2.1 Introduction . . . . .                                                    | 14        |
| 2.2 Limitations of Existing Models and Problem Statement . . . . .            | 14        |
| 2.3 Proposed Solution: The Adaptive Fusion Mechanism . . . . .                | 17        |
| 2.3.1 Formal Problem Definition . . . . .                                     | 17        |
| 2.3.2 The Adaptive Fusion Mechanism . . . . .                                 | 17        |
| 2.3.3 Intuition and Rationale . . . . .                                       | 18        |
| 2.4 Family of Proposed Hybrid Models . . . . .                                | 18        |
| 2.4.1 Hybrid BERT4Rec + LightGCN with Fixed Fusion . . . . .                  | 18        |
| 2.4.2 Hybrid BERT4Rec + LightGCN with Discrete Fusion . . . . .               | 18        |

|          |                                                                            |           |
|----------|----------------------------------------------------------------------------|-----------|
| 2.4.3    | Hybrid BERT4Rec + Temporal Graph Transformer (TGT) . . .                   | 19        |
| 2.4.4    | Hybrid BERT4Rec + Temporal Convolutional Network (TCN) .                   | 19        |
| 2.5      | Conclusion . . . . .                                                       | 20        |
| <b>3</b> | <b>Experiments and Results: Length-Adaptive Hybrid GNN-Transformer</b>     | <b>21</b> |
| 3.1      | Introduction . . . . .                                                     | 21        |
| 3.2      | Datasets . . . . .                                                         | 22        |
| 3.3      | Preprocessing . . . . .                                                    | 22        |
| 3.4      | Hyperparameters . . . . .                                                  | 24        |
| 3.5      | Baselines . . . . .                                                        | 25        |
| 3.6      | Main Results Tables . . . . .                                              | 28        |
| 3.7      | Experimental Results . . . . .                                             | 28        |
| 3.7.1    | Overall Test Metrics . . . . .                                             | 29        |
| 3.7.2    | Performance by History Length . . . . .                                    | 30        |
| 3.7.3    | Generalization Analysis . . . . .                                          | 31        |
| 3.7.4    | Summary . . . . .                                                          | 36        |
| 3.8      | Ablation Study . . . . .                                                   | 36        |
| 3.9      | Conclusion . . . . .                                                       | 39        |
| <b>4</b> | <b>Item Description Strategies for LLM-Based Sequential Recommendation</b> | <b>40</b> |
| 4.1      | Introduction . . . . .                                                     | 40        |
| 4.2      | The LLM2Sequential Pipeline . . . . .                                      | 40        |
| 4.3      | Item Description Strategies . . . . .                                      | 41        |
| 4.4      | Conclusion . . . . .                                                       | 43        |
| <b>5</b> | <b>Experiments and Results: PromptCraft-SeqRec</b>                         | <b>45</b> |
| 5.1      | Introduction . . . . .                                                     | 45        |
| 5.2      | Dataset and Experimental Setup . . . . .                                   | 45        |
| 5.3      | Recommendation Results . . . . .                                           | 47        |
| 5.4      | Embedding Quality Analysis . . . . .                                       | 49        |
| 5.5      | Comparison with the Published LLM2Sequential Baseline . . . . .            | 51        |
| 5.6      | Discussion . . . . .                                                       | 52        |
| 5.7      | Conclusion . . . . .                                                       | 53        |
|          | <b>Conclusion Générale et Perspectives</b>                                 | <b>54</b> |
|          | <b>Productions Scientifiques</b>                                           | <b>56</b> |
|          | <b>General Conclusion and Future Work</b>                                  | <b>57</b> |



## List of Figures

|     |                                                                                                                                                                                                                                                                                                                                     |    |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1.1 | Taxonomy of Recommendation System Types. . . . .                                                                                                                                                                                                                                                                                    | 5  |
| 1.2 | Illustration of a user interaction timeline for next-item prediction. . . .                                                                                                                                                                                                                                                         | 6  |
| 1.3 | Architecture illustration of a sequence-based model such as SASRec or BERT4Rec. . . . .                                                                                                                                                                                                                                             | 8  |
| 1.4 | Illustration of a user-item bipartite graph and embedding propagation in LightGCN. . . . .                                                                                                                                                                                                                                          | 9  |
| 1.5 | Pipeline for using item descriptions to generate embeddings with an LLM and feed them into a recommender. . . . .                                                                                                                                                                                                                   | 10 |
| 2.1 | Comparison of sparse and dense user history profiles and their ideal information sources. . . . .                                                                                                                                                                                                                                   | 16 |
| 3.1 | Preprocessing pipeline applied to both datasets. . . . .                                                                                                                                                                                                                                                                            | 23 |
| 3.2 | Test metrics heatmap on ML-1M. Darker cells indicate higher values. Hybrid models are prefixed with BERT-Hybrid. . . . .                                                                                                                                                                                                            | 29 |
| 3.3 | Test metrics heatmap on ML-100K. Darker cells indicate higher values. . . .                                                                                                                                                                                                                                                         | 29 |
| 3.4 | NDCG@10 broken down by user history length group on ML-1M. Short: $n_u \leq 10$ ; Medium: $10 < n_u \leq 30$ ; Overall: all users. . . . .                                                                                                                                                                                          | 30 |
| 3.5 | NDCG@10 broken down by user history length group on ML-100K. . . .                                                                                                                                                                                                                                                                  | 31 |
| 3.6 | $\Delta\text{NDCG}@10$ (Val – Test) per epoch on ML-1M for all four hybrid models. The green band marks the healthy range ( $\pm 0.01$ ). Red dots indicate epochs where the gap exceeds the threshold. . . . .                                                                                                                     | 32 |
| 3.7 | $\Delta\text{NDCG}@10$ (Val – Test) per epoch on ML-100K for all four hybrid models. The green band marks the healthy range ( $\pm 0.01$ ). Red dots indicate epochs where the gap exceeds the threshold. . . . .                                                                                                                   | 35 |
| 5.1 | Certificate of participation for the paper "Dynamic Sequence Modeling via Hybrid GNN-Transformer Architecture for Robust Sequential Recommendation" presented at the 2nd National Conference on Artificial Intelligence and Digital Transition (CAIDT'26), University of Kasdi Merbah, Ouargla, Algeria, April 29–30, 2026. . . . . | 56 |



## List of Tables

|     |                                                                                                                                                                                                             |    |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1.1 | Representative sequence-based recommendation models. . . . .                                                                                                                                                | 8  |
| 1.2 | Representative graph-based recommendation models. . . . .                                                                                                                                                   | 9  |
| 1.3 | Examples of how LLMs are used in sequential recommendation. . . . .                                                                                                                                         | 11 |
| 1.4 | Ranking Metrics for Sequential Recommendation. . . . .                                                                                                                                                      | 12 |
| 2.1 | Overview of the four proposed hybrid model variants. . . . .                                                                                                                                                | 19 |
| 3.1 | Statistics of the datasets used in this chapter after preprocessing. . . . .                                                                                                                                | 22 |
| 3.2 | Hyperparameter settings for all models. . . . .                                                                                                                                                             | 26 |
| 3.3 | Summary of baseline models used in this chapter. . . . .                                                                                                                                                    | 27 |
| 3.4 | Test performance on MovieLens-1M. Best results bold, second-best underlined. • = proposed hybrid. . . . .                                                                                                   | 28 |
| 3.5 | Test performance on MovieLens-100K. Best results bold, second-best underlined. • = proposed hybrid. . . . .                                                                                                 | 28 |
| 3.6 | Ablation study on ML-1M (HR@10 / NDCG@10 / MRR@10). . . . .                                                                                                                                                 | 37 |
| 3.7 | Model efficiency and statistical significance on ML-1M. • = proposed hybrid. * $p < 0.05$ . . . . .                                                                                                         | 38 |
| 4.1 | The seven item description strategies with illustrative examples. The example item is a moisturizing cream from the Amazon Beauty dataset. . . . .                                                          | 44 |
| 5.1 | Statistics of the Amazon Beauty dataset after 5-core filtering. Sparsity is computed as $1 - \frac{\#Interactions}{\#Users \times \#Items}$ . . . . .                                                       | 46 |
| 5.2 | Recommendation results on Amazon Beauty for all description strategies. SASRec hyperparameters are held constant across all runs. $\Delta$ NDCG@10 is relative to the T1 baseline. . . . .                  | 47 |
| 5.3 | Intrinsic embedding quality metrics for each description strategy. . . . .                                                                                                                                  | 50 |
| 5.4 | Comparison of our best strategies against the published LLM2Sequential baseline (Boz et al., 2025) on Amazon Beauty. $\Delta$ values are relative to the Boz et al., (2025) title-only result (T1). . . . . | 51 |

# General Introduction

## Context and Motivation

The era of digital transformation has connected millions of users through online platforms that have become central to daily life. E-commerce sites allow people to purchase and sell products worldwide, while streaming services such as YouTube provide instant access to entertainment including films, music, and video content. The rapid developments in the digital domain have produced an exponential growth in the volume of online content. Such an abundance of choices makes it increasingly difficult for users to locate relevant content aligned with their interests, and can impair or paralyse the decision-making process. Recommendation systems were developed specifically to address this challenge. By learning user interests and behavioral patterns, they help users discover products, services, or content they are likely to enjoy. Recommendation systems span several categories; sequential recommendation is one of the most active research directions in recent years. Unlike classical collaborative filtering approaches, which identify user interests through group similarities, sequential recommendation models exploit the temporal order of user interaction histories to predict the next item a user is likely to engage with - whether through purchase, viewing, or listening. By capturing the ordered sequence of past behaviors, these models better track dynamic user interests and provide more timely recommendations.

Deep learning has been a major driving force in sequential recommendation. Among early contributions, [Hidasi et al., \(2016\)](#) showed that gated recurrent networks are effective at capturing session-level dynamics for next-item prediction. Attention mechanisms and Transformer-based architectures later provided a substantial leap forward: models such as SASRec ([Kang and McAuley, 2018](#)) and BERT4Rec ([Sun et al., 2019](#)) demonstrated that learning dependencies across entire interaction sequences yields better accuracy and scalability than recurrent approaches. Graph Neural Networks were subsequently introduced to model the global relational structure among users and items, extracting collaborative information that purely sequential models cannot access.

However, these two paradigms have complementary strengths and weaknesses. Graph Neural Networks excel at capturing global item co-occurrence patterns but are limited in modeling the fine-grained temporal context within a user’s history. Sequence-based Transformer models are strong at encoding recent user preferences but cannot leverage

information beyond an individual user’s own interactions. Existing hybrid approaches generally combine these two components using a fixed, user-agnostic fusion weight. Yet users differ substantially in how much interaction history they have accumulated, and the relative reliability of the graph signal versus the sequential signal varies accordingly. Users with sparse histories benefit more from global collaborative information, while users with rich histories can better exploit detailed sequential patterns.

At the same time, Large Language Models (LLMs) such as GPT, LLaMA, and BERT, pre-trained on large text corpora, have demonstrated a remarkable capacity to encode semantic knowledge about items and contexts (L. Wu et al., 2023; Liu et al., 2024b). Their integration into recommendation systems has opened a new direction: using LLM-derived embeddings to enrich item representations with semantic information that interaction data alone cannot provide (Harte et al., 2023; Liu et al., 2024a). However, a fundamental question remains underexplored: how should the textual description of an item be constructed in order to maximally leverage the semantic content captured by LLM embeddings in a sequential recommendation setting? Different description strategies produce embeddings with different properties, which may significantly influence downstream recommendation quality.

## Addressed Research Problems

This memoir addresses two complementary research problems:

**Problematic 1** - *How to dynamically adapt the fusion between global graph signals and local sequential modelling according to each user’s interaction history profile?*

**Problematic 2** - *How does the choice of item textual description strategy influence the quality of LLM embeddings in sequential recommendation?*

## Main Contributions

This work presents two main contributions:

- **Contribution 1:** A hybrid GNN-Transformer model with adaptive fusion based on user interaction history length.
- **Contribution 2:** A systematic study of the impact of item description strategies on LLM embedding quality for sequential recommendation.

# Memoir Organization

The rest of this memoir is structured as follows:

**Chapter 1** describes the basic concepts and the state-of-the-art methods for sequential recommendation systems. It covers classical approaches, sequence-based and graph-based deep learning models, Large Language Models, and evaluation criteria. It ends with the research gaps that motivated this work.

**Chapter 2** proposes our first contribution: a Length-Adaptive Hybrid GNN-Transformer. It describes the shortcomings of previous models, formalizes the problem, and provides the details of the proposed architecture and fusion method.

**Chapter 3** illustrates the experiments of the first contribution. It contains the description of datasets, pre-processing, baselines, parameters, experimental results, and an ablation study.

**Chapter 4** discusses the second contribution: a comprehensive analysis of item description strategies for LLM-based sequential recommendation. It presents the **LLM2Sequential** pipeline and seven description strategies proposed in this work.

**Chapter 5** presents the evaluation experiments of the second contribution and compares all the strategies against a previous work ([Boz et al., 2025](#)).

Lastly, a **General Conclusion** presents the main findings, the limitations, and future research directions.

# Foundations and State of the Art

## 1.1 Introduction to Recommendation Systems

Recommendation systems (RS) are methods for filtering large volumes of data and surfacing items that match a given user's tastes and interests ([Castells and Jannach, 2023](#)). As digital catalogs grow - from product listings on e-commerce platforms to the libraries of streaming services - the volume of available choices has outpaced users' capacity to evaluate them individually, creating what the literature commonly refers to as the information overload problem ([Raza et al., 2024](#)). The goal of RS is to identify item patterns according to item characteristics and user interest profiles so as to recommend relevant content by analysing behavioral signals. Recommendation systems have become core infrastructure in modern intelligent applications across domains including e-commerce, media, and music ([Raza et al., 2024](#)).

There are three common paradigms used to develop a recommendation system, as illustrated in Figure 1.1:

- **Collaborative Filtering (CF):** Recommends items by leveraging the past behavior of users, on the assumption that users who agreed in the past will continue to agree in the future. CF can either use the interaction data itself (memory-based) or latent factor models learnt from the data (model-based) ([Castells and Jannach, 2023](#)).
- **Content-Based Filtering (CB):** Recommends items with the same characteristics as the ones the user previously showed affinity for, by comparing item features with an individual user profile ([Raza et al., 2024](#)).
- **Hybrid Systems:** Combinations of CF and CB methods are often developed in order to overcome the drawbacks of both methods - for example cold-start in CF and over-specialization in CB - and therefore build more efficient recommendation systems ([Raza et al., 2024](#)).

Out of the above, sequential recommendation systems form a subcategory and have gradually become a more and more important area of collaborative filtering, where the order in which users interacted with items is considered. This work concentrates only on this type of systems. Formally they are defined below.

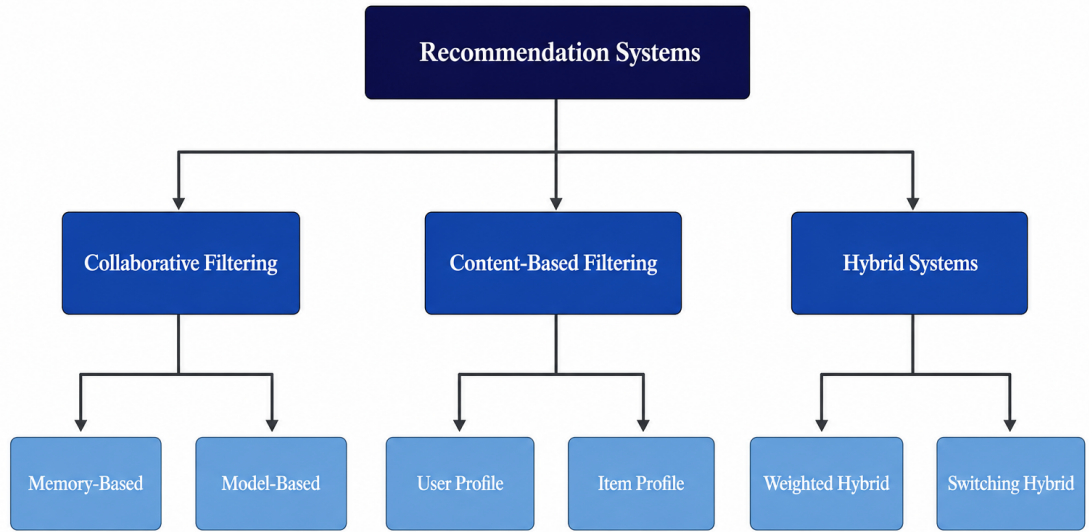


Figure 1.1: Taxonomy of Recommendation System Types.

## 1.2 Sequential Recommendation

In sequential recommendation (SR), the task of predicting the next item to recommend to a user depends not only on the set of items with which the user has interacted, but also on the specific order of these interactions (Pan et al., 2024). This sequence is important, as users’ preferences are dynamic: short-term preferences often have higher weight than long-term ones and the meaning of the action is interpreted differently based on the context that precedes it (for example, clicking an item does not carry the same semantics as purchasing it). SR is typically framed as the task of predicting the next item given a sequence of past interactions and is naturally suited for scenarios where dynamic, short-term intent needs to be modeled, such as e-commerce, news recommendation, or media streaming (Hidasi et al., 2016; Kang and McAuley, 2018; Sun et al., 2019).

Deep learning has led to major advances in the area of SR, due to models that are able to incorporate sequential dependencies explicitly. Among the early ones, the use of gated recurrent networks such as GRU4Rec was shown to be effective in capturing session-level dynamics and improving next-item prediction (Hidasi et al., 2016). Self-attention based models, like SASRec, proposed an effective solution to model dependencies across full interaction sequences (Kang and McAuley, 2018), and then BERT4Rec utilized bidirectional context through a masked item prediction objective (Sun et al., 2019).

These papers have consolidated SR as a major research direction in recommender systems.

In practice, SR remains challenging because interaction sequences can be very sparse, short, long, or heterogeneous, which complicates accurate modeling. Short sequences contain little evidence about user preferences, while long sequences may include irrelevant interactions that dilute the user’s current intent. Moreover, variable time gaps between interactions and differing action semantics increase the difficulty of balancing short-term and long-term interests. Consequently, recent work emphasizes more powerful sequence encoders, hybrid architectures that combine global and local signals, incorporation of side information, and integration with large language models to improve robustness and representation quality (Pan et al., 2024).

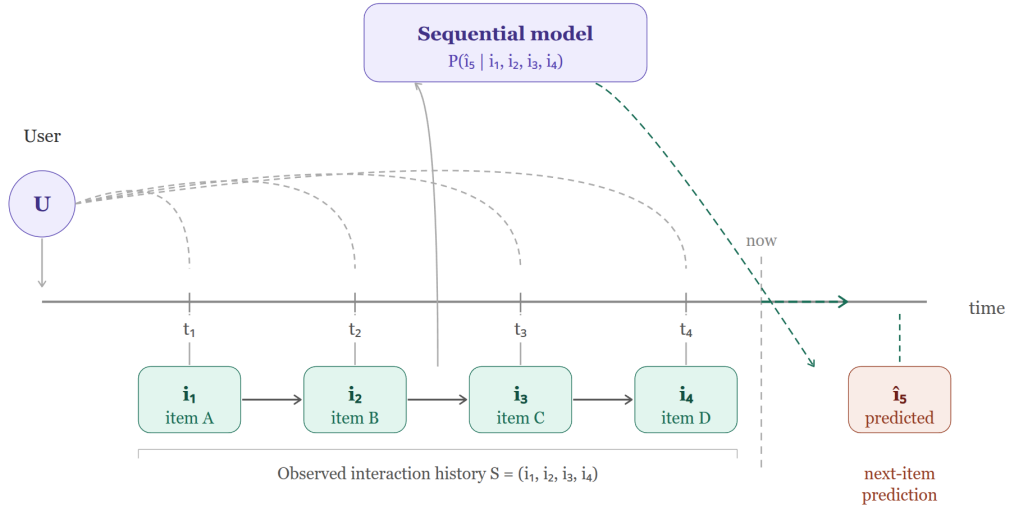


Figure 1.2: Illustration of a user interaction timeline for next-item prediction.

### 1.3 Sequence-Based Models

Sequence-based models constitute an important class of sequential recommendation methods that explicitly exploit the ordered nature of user interaction histories. Unlike simpler approaches that treat a user’s past actions as an unordered set, these models learn patterns from the exact sequence of events in order to predict the next likely item. This family of methods formed one of the earliest and most influential directions in sequential recommendation, and it includes recurrent, convolutional, and Transformer-based architectures.

### 1.3.1 GRU4Rec

GRU4Rec was one of the first deep learning models proposed for session-based recommendation and showed that recurrent neural networks can effectively model sequential user behavior (Hidasi et al., 2016). The model uses gated recurrent units to process interaction sequences in order, allowing it to capture short-term dependencies and session-level dynamics. Its key idea is that the hidden state summarizes the recent behavior of the user, which is then used to predict the next item in the sequence. This made GRU4Rec an important baseline for later sequential recommendation research.

### 1.3.2 Caser

Caser introduces a convolutional approach to sequential recommendation by embedding a user’s recent interaction history into a two-dimensional representation and applying convolutional filters over it (Tang and Wang, 2018). The model is designed to capture both general user preferences and local sequential patterns through vertical and horizontal convolutions. In contrast to recurrent models, Caser views the sequence as a structured pattern detection problem, which allows it to learn influential subsequences and short-range dependencies. This makes Caser a representative CNN-based baseline in the SR literature.

### 1.3.3 SASRec

SASRec applies self-attention to sequential recommendation, using a Transformer-style architecture to model dependencies between interacted items across the full interaction history (Kang and McAuley, 2018). Instead of processing the sequence strictly step by step, the model learns which past items are most relevant to the current prediction through attention weights. This gives SASRec a strong ability to capture both short- and long-term behavioral patterns while remaining efficient to train. It became one of the most influential Transformer-based baselines in sequential recommendation.

### 1.3.4 BERT4Rec

BERT4Rec extends the Transformer idea by adopting a bidirectional encoding strategy and a masked item prediction objective (Sun et al., 2019). Rather than predicting the next item in a strictly left-to-right fashion, the model randomly masks items in the sequence and learns to reconstruct them from both left and right context. This training objective encourages richer sequence representations and often improves the quality of learned user behavior embeddings. BERT4Rec is widely used as a strong benchmark for sequential recommendation due to its ability to model contextual dependencies more deeply.



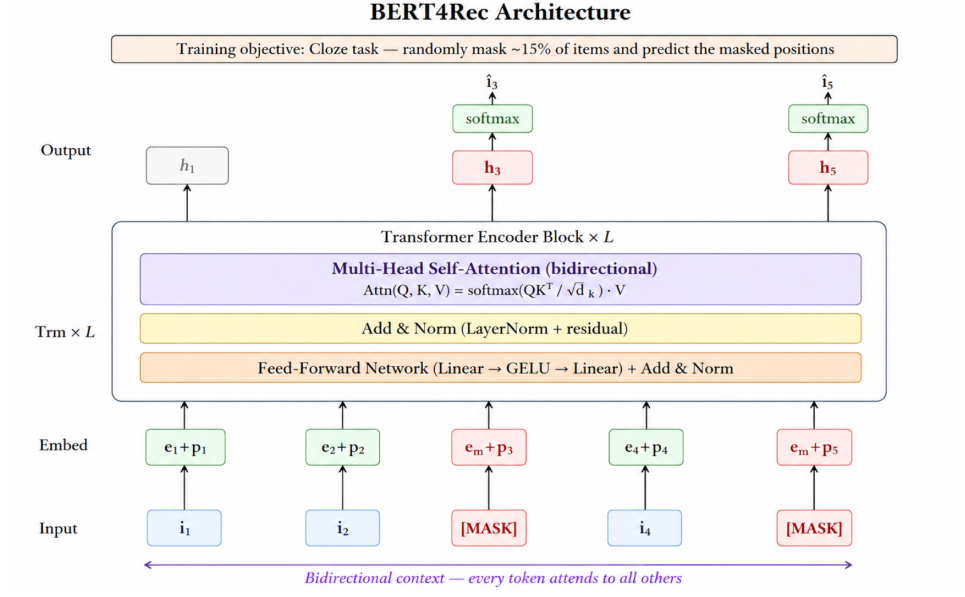


Figure 1.3: Architecture illustration of a sequence-based model such as SASRec or BERT4Rec.

| Model    | Approach                             | Key idea                                                                          |
|----------|--------------------------------------|-----------------------------------------------------------------------------------|
| GRU4Rec  | Recurrent neural network             | Models the interaction sequence step by step with gated recurrent units.          |
| Caser    | Convolutional neural network         | Learns local sequential patterns from a sequence-as-image representation.         |
| SASRec   | Self-attention / Transformer encoder | Uses attention to focus on the most relevant past items for next-item prediction. |
| BERT4Rec | Bidirectional Transformer encoder    | Learns contextual sequence representations using masked item prediction.          |

Table 1.1: Representative sequence-based recommendation models.

## 1.4 Graph-Based Models / GNN

Graph-based recommender systems model user-item interactions as a graph, where users and items are represented as nodes and historical interactions are encoded as edges. This formulation is useful because recommendation data are naturally relational: a user’s preference can be inferred not only from their own history, but also from their position in the interaction graph and from the neighborhoods of connected users and items. Graph neural networks therefore provide a powerful way to propagate collaborative signals, capture higher-order connectivity, and learn richer embeddings than methods based solely on local sequence patterns (S. Wu et al., 2020; He et al., 2020).

### 1.4.1 LightGCN

LightGCN is one of the most influential graph-based recommendation models because it simplifies graph convolution to its essential component: neighborhood aggregation (He et al., 2020). Instead of using feature transformation and nonlinear activation, LightGCN directly propagates user and item embeddings over the bipartite interaction graph and combines the embeddings from multiple propagation layers to obtain the final representation. This makes the model both efficient and effective, and it has become a standard baseline for graph-based collaborative filtering. Its main strength is the ability to capture collaborative signals from the full interaction graph while remaining simple enough to train and interpret.

| Model    | Approach                      | Key idea                                                                                                                     |
|----------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| LightGCN | Graph collaborative filtering | Learns user and item embeddings by propagating them over the bipartite interaction graph with only neighborhood aggregation. |

Table 1.2: Representative graph-based recommendation models.

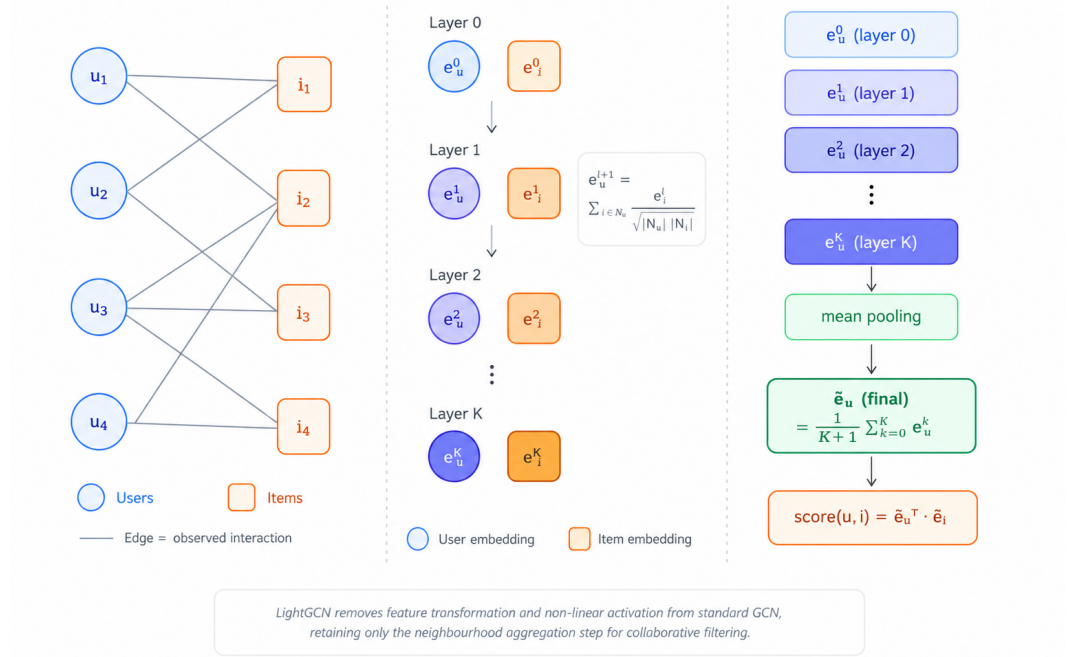


Figure 1.4: Illustration of a user-item bipartite graph and embedding propagation in LightGCN.

## 1.5 LLMs Applied to Sequential Recommendation

Large Language Models (LLMs) have recently entered recommendation systems by bringing strong semantic understanding of text, contextual reasoning, and transferable representations learned from large-scale corpora. In sequential recommendation, this is particularly valuable because many items are described by textual metadata such as titles, descriptions, reviews, or categorical attributes. Rather than relying only on item identifiers or interaction history, LLM-based methods can transform item text into dense embeddings that encode semantic similarity and can be injected into recommendation pipelines as initialization, side information, or direct ranking signals (Harte et al., 2023; Liu et al., 2024a; L. Wu et al., 2023; Liu et al., 2024b).

A central idea in this line of work is that the textual description of an item can act as a semantic bridge between sparse interaction data and richer domain knowledge. When item descriptions are encoded by an LLM, the resulting embeddings can capture not only surface-level wording, but also relations between concepts, product properties, and user-facing meaning. These embeddings can then be used to improve sequential recommenders by initializing item representations, enhancing content-based signals, or supporting hybrid models that combine textual and behavioral information. This is especially helpful in cold-start or sparse-data settings, where item IDs alone provide little information about the item’s semantics.

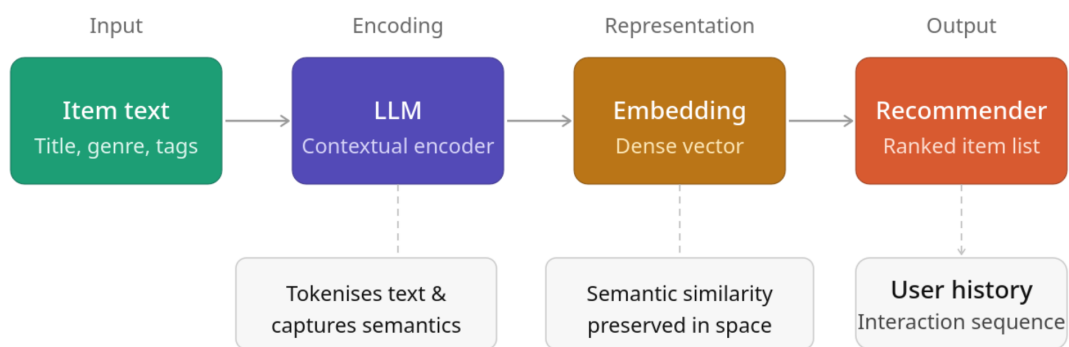


Figure 1.5: Pipeline for using item descriptions to generate embeddings with an LLM and feed them into a recommender.

Recent studies have shown that LLMs can be used in several ways for sequential recommendation. Some approaches initialize strong sequential models such as BERT4Rec

or SASRec with LLM-derived embeddings, while others fine-tune the LLM directly for recommendation or use it as a semantic encoder for items and user profiles (Harte et al., 2023; Liu et al., 2024a; L. Wu et al., 2023). Survey work also highlights a broader shift from treating LLMs only as feature extractors toward using them as integrated components of recommendation pipelines, including knowledge enhancement, interaction enhancement, and model enhancement (Liu et al., 2024b). This makes LLMs a promising direction for improving both representation quality and generalization in sequential recommendation.

At the same time, the usefulness of LLMs depends strongly on how item text is constructed and presented. A short product title, a structured attribute list, and a longer user-centric description can lead to different embedding spaces and therefore different recommendation behaviors. For this reason, item text design is not a trivial preprocessing step, but an important modeling choice that can influence semantic richness, distinctiveness, and downstream performance. This observation motivates the empirical study carried out in this memoir, where different item description strategies are analyzed in the context of sequential recommendation.

| Use case                 | Method                         | Key idea                                                                 |
|--------------------------|--------------------------------|--------------------------------------------------------------------------|
| Embedding initialization | LLM $\rightarrow$ item vectors | Use semantic embeddings to initialize or enrich sequential recommenders. |
| Direct recommendation    | LLM as recommender             | Fine-tune or prompt the LLM to rank candidate items directly.            |
| Hybrid enhancement       | LLM + sequential model         | Combine textual embeddings with behavioral sequence modeling.            |

Table 1.3: Examples of how LLMs are used in sequential recommendation.

## 1.6 Evaluation Metrics

To evaluate the performance of sequential recommendation models, this memoir uses three standard ranking-based metrics: Hit Ratio at  $K$  (HR@K), Normalized Discounted Cumulative Gain at  $K$  (NDCG@K), and Mean Reciprocal Rank (MRR). These metrics measure how well the model places the true next item among its top-ranked predictions. Since sequential recommendation is typically a top- $K$  ranking problem, such metrics are more informative than simple classification accuracy.

### 1.6.1 Hit Ratio at $K$

HR@K measures whether the ground-truth next item appears in the top- $K$  recommended items. It gives a binary reward of 1 if the target item is present in the rec-

ommendation list and 0 otherwise. As a result, HR@K captures the model’s ability to retrieve the correct item within a limited number of suggestions, but it does not consider the exact rank position inside the top- $K$ .

### 1.6.2 Normalized Discounted Cumulative Gain at $K$

NDCG@K evaluates both whether the correct item is retrieved and how high it is ranked in the recommendation list. Higher-ranked correct predictions receive larger scores, which makes NDCG@K more sensitive to the ordering quality of the top- $K$  list. This metric is especially useful when the position of the predicted item matters, which is often the case in practical recommendation scenarios.

### 1.6.3 Mean Reciprocal Rank

MRR measures the inverse of the rank position of the first relevant item. Unlike HR@K, it rewards models that place the target item very close to the top of the list. If the correct item is ranked first, the reciprocal rank is 1; if it is ranked second, it is  $1/2$ , and so on. MRR therefore reflects how early the desired item appears in the ranked output.

| Metric | Formula                                                                                                       | What it measures                                                          |
|--------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| HR@K   | $\text{HR@K} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\text{rank}_i \leq K)$                                     | Whether the true next item appears in the top- $K$ recommendations.       |
| NDCG@K | $\text{NDCG@K} = \frac{1}{N} \sum_{i=1}^N \frac{\mathbb{I}(\text{rank}_i \leq K)}{\log_2(\text{rank}_i + 1)}$ | Retrieval quality with a stronger reward for higher-ranked correct items. |
| MRR    | $\text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{\text{rank}_i}$                                               | How early the first relevant item appears in the ranked list.             |

Table 1.4: Ranking Metrics for Sequential Recommendation.

## 1.7 Conclusion

This chapter introduced the fundamental concepts underlying recommendation systems and provided an overview of the main paradigms used in practice, including collaborative filtering, sequence-based models, graph-based models, and LLM-enhanced approaches. It also presented the sequential recommendation setting, the principal baseline architectures used in this work, and the evaluation metrics employed to assess model performance.

Despite the progress made by these methods, several limitations remain, particularly in the way they model heterogeneous user histories, combine global and local signals, and exploit textual item information. These open questions motivate the research gaps addressed in the following chapter.

## 1.8 Identified Gaps and Positioning of Contributions

Despite the substantial progress achieved by recent sequential recommendation methods, two important gaps remain insufficiently addressed in the literature. First, many hybrid GNN-Transformer approaches combine global graph signals and local sequential signals using a fixed or uniform fusion strategy, without considering that users exhibit different interaction history profiles. In practice, users with sparse histories benefit more from global collaborative information, whereas users with rich histories can better exploit detailed sequential patterns. This observation motivates our first contribution, which introduces a length-adaptive fusion mechanism that dynamically balances graph-based and sequence-based representations according to the amount of available user history.

Second, although large language models have recently been integrated into recommendation systems, the influence of item textual description strategy on the quality of the resulting embeddings has not yet been systematically explored. Different forms of item text, such as titles, structured attributes, or user-oriented descriptions, may produce embeddings with different semantic properties and downstream effects on recommendation performance. This gap motivates our second contribution, which conducts a systematic study of how item description design affects LLM embeddings in sequential recommendation.

Together, these two gaps define the positioning of this memoir: the first contribution addresses the adaptive fusion of structural and sequential signals, while the second investigates the semantic role of item text in LLM-based sequential recommendation.

# Length-Adaptive Hybrid GNN-Transformer for Sequential Recommendation

## 2.1 Introduction

Sequential recommendation remains a challenging problem because user interaction histories are heterogeneous in both length and density. Transformer-based models such as SASRec (Kang and McAuley, 2018) and BERT4Rec (Sun et al., 2019) have demonstrated strong performance by capturing long-range dependencies within interaction sequences, but they rely on the availability of sufficiently long histories to generalize reliably. Graph Neural Network models such as LightGCN (He et al., 2020) address this by aggregating global collaborative signals from the full user-item interaction graph, making them more robust for sparse users. However, when used in isolation, each paradigm captures only one side of the recommendation signal: sequence models ignore global item co-occurrence structure, while graph models ignore the fine-grained temporal order of individual user interactions (S. Wu et al., 2020; Pan et al., 2024).

This chapter proposes a Length-Adaptive Hybrid GNN-Transformer, a model that combines both paradigms through a user-specific fusion mechanism controlled by the length of each user’s interaction history. The core idea is that the relative reliability of the graph signal versus the sequence signal is not constant across users: it varies directly with the amount of available sequential evidence. The proposed model quantifies this variation through a scalar weight  $\lambda(u)$  that dynamically adjusts the contribution of the GNN and Transformer branches for each user individually. The remainder of this chapter is organized as follows: Section 2.2 discusses the limitations of existing models and formally states the research problem; Section 2.3 introduces the adaptive fusion mechanism and the formal problem definition; Section 2.4 presents the family of hybrid model variants; and Section 2.5 concludes the chapter.

## 2.2 Limitations of Existing Models and Problem Statement

Despite the progress achieved by the models reviewed in Chapter 1, three concrete limitations can be identified in the existing literature on hybrid GNN-Transformer sequential recommendation.

**Limitation 1 - Fixed fusion weights in existing hybrid models.** Hybrid approaches that combine GNN and Transformer components, such as those surveyed in (S. Wu et al., 2020), typically apply a single global fusion weight that is shared uniformly across all users. This fixed weighting strategy treats every user identically, regardless of how much interaction history is available. As a result, the model cannot adapt to the fact that the relative informativeness of the graph signal versus the sequence signal varies substantially from one user to another. A fixed weight that is optimal on average will systematically underserve both sparse users and dense users (Pan et al., 2024).

**Limitation 2 - Transformer-only models fail sparse users.** Self-attention mechanisms, as used in SASRec (Kang and McAuley, 2018) and BERT4Rec (Sun et al., 2019), require a sufficient number of interactions to produce reliable contextualized representations. When a user has only a small number of past interactions, the self-attention layers lack enough context to generalize, and the resulting sequence embedding is noisy or underspecified. These models therefore provide poor recommendations for sparse and cold-start users, a critical segment in any real-world recommendation platform.

**Limitation 3 - GNN-only models waste dense users’ sequential patterns.** Graph-based models such as LightGCN (He et al., 2020) aggregate neighborhood information over the full interaction graph and produce strong collaborative embeddings for sparse users. However, they do not model the temporal order of a user’s individual interactions. For users with long, rich interaction histories, this means that fine-grained sequential patterns - the exact order in which items were consumed, recency effects, and evolving preferences - are entirely discarded. The graph signal alone is insufficient to capture dynamic intent for active users (Pan et al., 2024).

## Problem Statement

The three limitations above converge on a single fundamental question, which constitutes the first research problematic of this thesis:

*How to dynamically adapt the fusion between global graph signals and local sequential modelling according to each user’s interaction history profile?*

Answering this question requires a model that is aware of each user’s interaction density and uses this information to balance the GNN and Transformer contributions at the individual user level, rather than applying a fixed global weight. This is the design principle that motivates the proposed architecture described in the following section.



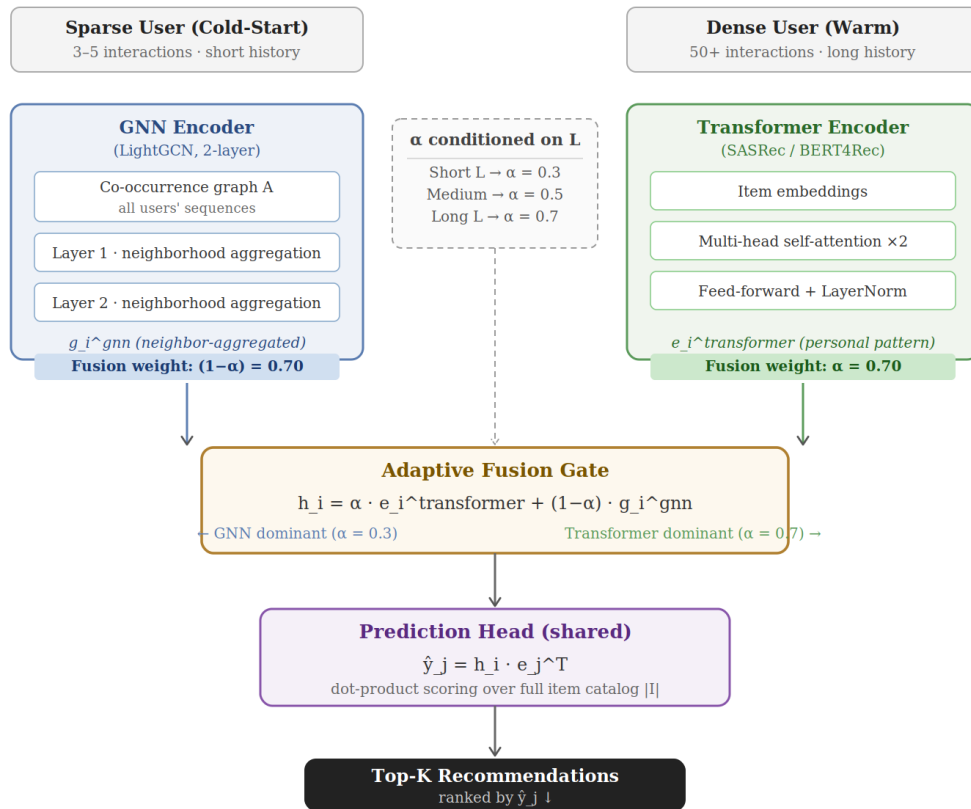


Figure 2.1: Comparison of sparse and dense user history profiles and their ideal information sources.

## 2.3 Proposed Solution: The Adaptive Fusion Mechanism

### 2.3.1 Formal Problem Definition

Let  $\mathcal{U}$  and  $\mathcal{I}$  denote the sets of users and items, respectively. For each user  $u \in \mathcal{U}$ , let  $S_u = (i_1, i_2, \dots, i_{n_u})$  be their interaction history of length  $n_u$ , where each  $i_t \in \mathcal{I}$  is an item consumed at time step  $t$ , ordered chronologically. The task of next-item sequential recommendation is defined as: given the prefix  $S_u^{(t)} = (i_1, \dots, i_t)$ , learn a scoring function

$$f_{\Theta}(u, S_u^{(t)}, j) \in \mathbb{R} \quad (2.1)$$

that assigns a relevance score to each candidate item  $j \in \mathcal{I}$ , where  $\Theta$  denotes all trainable parameters. At inference time, items are ranked by their scores and top- $K$  accuracy is evaluated via HR@ $K$ , NDCG@ $K$ , and MRR@ $K$  as defined in Chapter 1.

### 2.3.2 The Adaptive Fusion Mechanism

The proposed model computes two independent representations for each user  $u$ : a graph-based representation  $\mathbf{h}_u^{\text{GNN}}$ , produced by a Graph Neural Network encoder operating over the global item co-occurrence graph, and a sequence-based representation  $\mathbf{h}_u^{\text{Trans}}$ , produced by a Transformer encoder operating over the user's personal interaction history  $S_u$ . These two representations are then combined through a user-specific scalar weight  $\lambda(u) \in [0, 1]$ :

$$\mathbf{h}_u = \lambda(u) \cdot \mathbf{h}_u^{\text{Trans}} + (1 - \lambda(u)) \cdot \mathbf{h}_u^{\text{GNN}} \quad (2.2)$$

where the fusion weight  $\lambda(u)$  is derived from the user's interaction history length  $n_u$  and a normalization factor  $n_{\max}$ , the maximum history length observed in the training set:

$$\lambda(u) = \sigma\left(\frac{n_u}{n_{\max}}\right) \quad (2.3)$$

and  $\sigma(\cdot)$  denotes the sigmoid function  $\sigma(x) = 1/(1 + e^{-x})$ .

### 2.3.3 Intuition and Rationale

The intuition behind Equations 2.2 and 2.3 is straightforward. When a user has a short interaction history ( $n_u$  is small),  $\lambda(u)$  is close to 0, and the fused representation  $\mathbf{h}_u$  relies predominantly on  $\mathbf{h}_u^{\text{GNN}}$ . This is desirable because the GNN aggregates global co-occurrence signals from all users in the training set and is therefore informative even when individual sequence evidence is scarce. Conversely, when a user has a long interaction history ( $n_u$  is large),  $\lambda(u)$  approaches 1, and  $\mathbf{h}_u$  relies predominantly on  $\mathbf{h}_u^{\text{Trans}}$ . In this regime, the Transformer can exploit the full richness of the user’s personal sequential patterns, which carry more information than global averages.

The use of history length  $n_u$  as the proxy variable is motivated by the observation that the reliability of self-attention degrades when the input sequence is too short (Kang and McAuley, 2018; Sun et al., 2019), while GNN neighborhood aggregation is length-independent by construction (He et al., 2020). The sigmoid normalization in Equation 2.3 ensures that  $\lambda(u)$  remains bounded in  $[0, 1]$  and transitions smoothly between the two extremes, avoiding hard discontinuities in the embedding space.

## 2.4 Family of Proposed Hybrid Models

This work proposes four hybrid models, all built on the same core idea: combining a graph encoder with a sequential encoder, and blending their outputs based on how long a user’s interaction history is. The models share the same LightGCN (He et al., 2020) graph backbone and differ in how they encode sequences and how they control the blending.

### 2.4.1 Hybrid BERT4Rec + LightGCN with Fixed Fusion

The simplest variant applies a single fixed blending weight to all users equally. LightGCN provides collaborative signals from the item co-occurrence graph, while BERT4Rec (Sun et al., 2019) encodes the interaction sequence using bidirectional self-attention. The contribution of each encoder is set manually before training and never changes. This model serves as the baseline of the family: it confirms that combining the two encoders helps, but it does not adapt to differences between users.

### 2.4.2 Hybrid BERT4Rec + LightGCN with Discrete Fusion

The discrete variant introduces length-awareness by dividing users into three groups based on their history length: short, medium, and long. Each group receives a different fixed blending weight, giving short-history users more graph signal and

long-history users more sequential signal. The thresholds and weights are set manually and stay fixed during training. Compared to the fixed model, this variant is more principled because it acknowledges that different users need different encoder balances, while remaining fully interpretable and adding no trainable parameters.

### 2.4.3 Hybrid BERT4Rec + Temporal Graph Transformer (TGT)

This variant replaces LightGCN with a Temporal Graph Transformer, which extends standard graph attention with time-awareness. Concretely, it augments the attention keys with learned time encodings so that the model can weigh recent interactions more heavily than older ones when building item representations. BERT4Rec handles the sequential side as before. The two branches are fused through a single learnable gating parameter that is optimised during training, allowing the model to find the right balance automatically. This makes TGT-BERT4Rec the most expressive graph-based hybrid in the family, at the cost of higher complexity.

### 2.4.4 Hybrid BERT4Rec + Temporal Convolutional Network (TCN)

The TCN variant replaces the graph encoder entirely with a Temporal Convolutional Network. TCN uses stacked dilated causal convolutions, meaning it can capture long-range sequential dependencies without ever looking at future tokens. Because convolutions process sequences much faster than attention, this model is significantly more efficient than the transformer-only alternatives. The TCN and BERT4Rec branches share the same item embeddings and their outputs are combined through either a fixed weight, a learnable scalar, or a learned projection layer that concatenates both representations. This variant is particularly useful when inference speed or memory is a constraint.

Table 2.1: Overview of the four proposed hybrid model variants.

| Model           | Graph / Conv encoder       | Seq. encoder | Fusion                           |
|-----------------|----------------------------|--------------|----------------------------------|
| Fixed Fusion    | LightGCN                   | BERT4Rec     | Fixed scalar, same for all users |
| Discrete Fusion | LightGCN                   | BERT4Rec     | Fixed scalar per length group    |
| TGT-BERT4Rec    | Temporal Graph Transformer | BERT4Rec     | Learned gate                     |
| TCN-BERT4Rec    | Temporal Conv. Network     | BERT4Rec     | Fixed / learned / concat         |

## 2.5 Conclusion

This chapter proposed a Length-Adaptive Hybrid GNN-Transformer framework for sequential recommendation, built around the core idea that the optimal balance between global graph signals and local sequential signals is not constant but varies with each user’s interaction history length. The novelty of the proposed approach lies in the user-specific fusion weight  $\lambda(u)$ , which dynamically shifts the representation from GNN-dominated for sparse users toward Transformer-dominated for dense users, addressing a design flaw shared by all existing fixed-weight hybrid models (S. Wu et al., 2020; Pan et al., 2024). Two concrete variants were derived from this framework by combining LightGCN (He et al., 2020) with BERT4Rec (Sun et al., 2019), and using either a linear or sigmoid fusion schedule, allowing the effect of the fusion schedule choice to be studied independently. A current limitation of the framework is that  $\lambda(u)$  is computed from history length alone using a fixed formula, rather than being learned end-to-end from data, which may leave performance gains on the table for users whose optimal fusion point does not align perfectly with their history length. The experimental validation of both variants, including comparisons against the baseline models from Chapter 1 and a per-user-segment analysis, is presented in Chapter 3.

# Experiments and Results: Length-Adaptive Hybrid GNN-Transformer

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## 3.1 Introduction

Chapter 2 proposed a Length-Adaptive Hybrid GNN-Transformer framework for sequential recommendation, built around the idea that the relative contribution of global graph signals and local sequential signals should vary according to each user’s interaction history length. The framework was instantiated into two concrete model variants combining LightGCN (He et al., 2020) with BERT4Rec (Sun et al., 2019), using either a linear or sigmoid fusion schedule for  $\lambda(u)$ . This chapter provides the empirical validation of these two variants against a set of strong sequential and graph-based baselines on two standard benchmark datasets.

The experiments in this chapter are designed to answer three concrete questions. First, does the proposed adaptive fusion mechanism yield better overall recommendation performance compared to fixed-weight hybrid baselines and to single-paradigm models? Second, does the length-adaptive design specifically improve performance for sparse users, who are systematically underserved by Transformer-only models? Third, does each component of the proposed architecture - the GNN branch, the Transformer branch, and the adaptive  $\lambda(u)$  schedule - contribute independently to the final performance, or can any of them be removed without degrading results?

The remainder of this chapter is organized as follows. Section 3.2 describes the datasets used and justifies their selection. Section 3.3 details the preprocessing pipeline. Section 3.4 reports all hyperparameter settings. Section 3.5 presents the baseline models and the evaluation protocol. Section 3.7 reports and analyzes the main results. Section 3.8 presents the ablation study. Finally, Section 3.9 concludes the chapter and transitions to Chapter 4.

## 3.2 Datasets

The experiments are conducted on two standard sequential recommendation benchmarks derived from the MovieLens collection (Harper and Konstan, 2015), which are among the most widely used datasets in the sequential recommendation literature (Pan et al., 2024; Kang and McAuley, 2018; Sun et al., 2019). These datasets were chosen because they cover two different scales of interaction density, allowing the evaluation of model robustness across both sparse and dense user populations - a property that is particularly important for validating the length-adaptive design of the proposed model.

**MovieLens-100K (ML-100K)** contains 100,000 explicit ratings collected from 943 users on 1,682 movies. After filtering users with fewer than 5 interactions and converting ratings to implicit feedback (an interaction is recorded if and only if the rating is  $\geq 4$ ), the dataset yields 938 test users. This smaller dataset contains a higher proportion of sparse users relative to ML-1M, making it a valuable testbed for evaluating cold-start and sparse-user behavior.

**MovieLens-1M (ML-1M)** contains approximately 1 million ratings from 6,040 users on 3,706 movies. After the same preprocessing steps, the dataset yields 6,034 test users. Its larger scale and denser interaction distribution make it a complementary benchmark to ML-100K, allowing the evaluation of model scalability and performance under richer sequential evidence.

Table 3.1: Statistics of the datasets used in this chapter after preprocessing.

| Dataset | #Users | #Items | #Interactions | Sparsity | Avg. seq. length |
|---------|--------|--------|---------------|----------|------------------|
| ML-100K | 943    | 1,682  | 100,000       | 93.70%   | 106.0            |
| ML-1M   | 6,040  | 3,706  | 1,000,209     | 95.53%   | 165.5            |

Sparsity is computed as  $1 - \frac{\#Interactions}{\#Users \times \#Items}$ . Both datasets confirm the typical sparsity challenge in collaborative filtering, motivating the use of GNN signals to supplement sequential evidence for users with short histories.

## 3.3 Preprocessing

The raw MovieLens datasets are preprocessed following the standard pipeline used in the sequential recommendation literature (Kang and McAuley, 2018; Sun et al., 2019; Pan et al., 2024). The preprocessing steps are applied identically to both ML-100K and ML-1M to ensure fair comparison across datasets. The full pipeline is illustrated

in Figure 3.1.

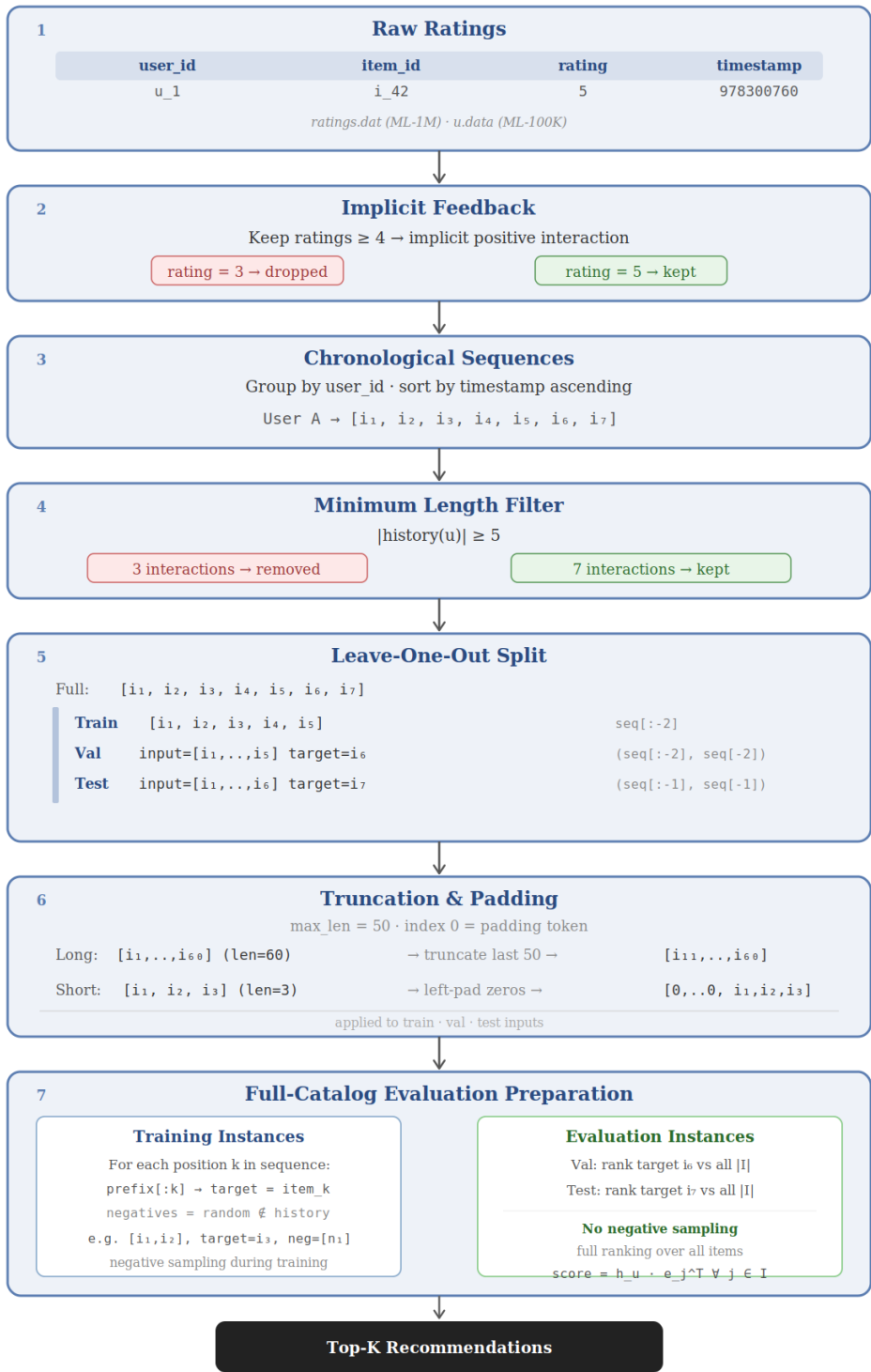


Figure 3.1: Preprocessing pipeline applied to both datasets.



**Step 1 - Implicit feedback conversion.** All explicit ratings are converted to implicit feedback: an interaction is recorded if and only if the rating value is greater than or equal to 4. This binary conversion is standard in sequential recommendation, where the goal is to model the order of consumed items rather than preference intensity.

**Step 2 - 5-core filtering.** Users and items with fewer than 5 interactions are removed from the dataset. This filtering step ensures that every user has enough interaction history to construct a meaningful training sequence, and that every item appears sufficiently often to learn a reliable embedding. The filter is applied iteratively until no user or item falls below the threshold.

**Step 3 - Chronological sorting.** For each user, all interactions are sorted in ascending order of timestamp. This produces a chronologically ordered interaction sequence  $S_u = (i_1, i_2, \dots, i_{n_u})$  for each user  $u$ , which is the input format required by all sequential models.

**Step 4 - Leave-one-out split.** The dataset is split into training, validation, and test sets following the leave-one-out protocol (Kang and McAuley, 2018). For each user, the last item in the interaction sequence is held out for testing, the second-to-last item is held out for validation, and all remaining items form the training set. This protocol is standard in the sequential recommendation literature and ensures that the temporal order of interactions is preserved.

**Step 5 - Sequence truncation.** To manage computational cost, interaction sequences are truncated to a maximum length of 200 for hybrid models and 50 for pure sequential baselines. When a sequence exceeds the maximum length, the most recent interactions are retained and earlier ones are discarded, preserving the most informative recent context.

**Step 6 - Negative sampling for evaluation.** During evaluation, each positive test item is ranked against all items in the full item catalog. No negative sampling is applied at test time, ensuring that the reported metrics reflect full-catalog ranking performance rather than sampled approximations. During training, 100 negative items are sampled uniformly at random per positive interaction for the binary cross-entropy loss.

## 3.4 Hyperparameters

All models are implemented and trained under identical conditions to ensure fair comparison. The hyperparameters used in this chapter are reported in Table 3.2.

General architecture hyperparameters - embedding dimension, number of attention heads, number of Transformer blocks, and dropout rate - are fixed to the same values across all models. Training hyperparameters - optimizer, learning rate, batch size, and early stopping criterion - are also held constant. This design ensures that any observed performance differences between models are attributable to architectural choices rather than training configuration differences.

The GNN-specific hyperparameters - number of GCN layers and co-occurrence graph construction parameters - are fixed based on standard settings from the LightGCN literature (He et al., 2020). The co-occurrence window size  $w = 5$  and minimum co-occurrence threshold  $\tau = 2$  were chosen to balance graph density and edge quality: a window that is too narrow misses meaningful item relationships, while a threshold that is too low introduces noisy edges that degrade embedding quality.

For the HybridDiscrete variant, the three fusion weight values  $\alpha_{\text{short}}$ ,  $\alpha_{\text{mid}}$ , and  $\alpha_{\text{long}}$  are treated as hyperparameters tuned on the validation set using NDCG@10 as the selection criterion. The history length thresholds  $L_{\text{short}}^* = 10$  and  $L_{\text{long}}^* = 30$  partition users into three bins and were selected based on the empirical distribution of history lengths in the training set.

## 3.5 Baselines

The proposed hybrid model variants are evaluated against five baseline models grouped into three categories: sequential-only models, a graph-only model, and hybrid ablation variants derived from the proposed framework itself. All models are evaluated using the leave-one-out protocol described in Section 3.3, where each positive test item is ranked against all items in the full catalog. Performance is reported at cut-off  $K \in \{5, 10, 20\}$  for HR@ $K$ , NDCG@ $K$ , and MRR@ $K$ .

### Sequential-Only Baselines

**GRU4Rec** (Hidasi et al., 2016) is a recurrent neural network model that uses gated recurrent units to process interaction sequences step by step, capturing short-term session-level dynamics. It serves as the recurrent baseline and represents the earliest class of deep sequential recommendation models.

**SASRec** (Kang and McAuley, 2018) applies unidirectional causal self-attention to model long-range dependencies in interaction sequences. It is trained with binary cross-entropy loss and is one of the strongest and most widely adopted Transformer-based baselines in sequential recommendation.

Table 3.2: Hyperparameter settings for all models.

| Hyperparameter                                                                         | Value | Search range / Note     |
|----------------------------------------------------------------------------------------|-------|-------------------------|
| <i>General</i>                                                                         |       |                         |
| Embedding dimension $d$                                                                | 64    | Fixed                   |
| Attention heads                                                                        | 2     | Fixed                   |
| Transformer blocks $B$                                                                 | 2     | Fixed                   |
| Dropout rate                                                                           | 0.2   | Fixed                   |
| Max sequence length (hybrid)                                                           | 200   | Fixed                   |
| Max sequence length (base-lines)                                                       | 50    | Fixed                   |
| <i>Training</i>                                                                        |       |                         |
| Optimizer                                                                              | Adam  | Fixed                   |
| Learning rate $\eta$                                                                   | 0.001 | Fixed                   |
| Batch size                                                                             | 256   | Fixed                   |
| Max epochs                                                                             | 200   | Fixed                   |
| Early stopping patience                                                                | 20    | Validation NDCG@10      |
| Negatives per positive                                                                 | 100   | Fixed                   |
| <i>GNN Encoder</i>                                                                     |       |                         |
| GCN layers $L_{\text{GNN}}$                                                            | 2     | Fixed                   |
| Co-occurrence window $w$                                                               | 5     | Fixed                   |
| Min co-occurrence threshold $\tau$                                                     | 2     | Fixed                   |
| <i>HybridDiscrete Fusion</i>                                                           |       |                         |
| Short-history threshold $L_{\text{short}}^*$                                           | 10    | Fixed                   |
| Long-history threshold $L_{\text{long}}^*$                                             | 30    | Fixed                   |
| $\alpha_{\text{short}}$                                                                | 0.3   | Tuned on validation set |
| $\alpha_{\text{mid}}$                                                                  | 0.5   | Tuned on validation set |
| $\alpha_{\text{long}}$                                                                 | 0.7   | Tuned on validation set |
| *TCN-BERT4Rec and TGT-BERT4Rec use the same general hyperparameters (embedding dim 64) |       |                         |

**BERT4Rec** (Sun et al., 2019) uses bidirectional self-attention and a masked item prediction objective to produce rich contextual sequence representations. It is the sequential backbone shared by all proposed hybrid variants and serves as the primary baseline against which the hybrid framework is evaluated.

**Caser** (Tang and Wang, 2018) applies horizontal and vertical convolutional filters over a sequence-as-image representation to capture both point-level and union-level sequential patterns. It represents the convolutional family of sequential models.

## Graph-Only Baseline

**LightGCN** (He et al., 2020) simplifies graph convolution to pure neighborhood aggregation over the user-item bipartite graph, without feature transformation or nonlinear activation. It is included not as a competitive sequential baseline but as a

reference point that isolates the contribution of pure collaborative filtering signals in a next-item prediction setting. Its relatively low performance on sequential metrics directly motivates the hybrid design: static graph-based collaborative filtering alone is insufficient when temporal ordering matters.

## Hybrid Ablation Variants

The following three variants of the proposed framework are included as internal ablation baselines to isolate the contribution of each architectural component:

**HybridFixed** uses a single global fusion weight  $\alpha$  shared uniformly across all users, trained end-to-end as a learnable scalar. It verifies that combining GNN and Transformer embeddings is beneficial, while treating every user identically regardless of history length.

**TCN-BERT4Rec** replaces the GCN graph encoder with a dilated causal convolutional network that encodes local sequential transition patterns. It tests whether local convolutional features provide complementary information to the global self-attention of BERT4Rec, without relying on a pre-built item graph.

**TGT-BERT4Rec** augments the item co-occurrence graph with temporal information by encoding interaction timestamps as continuous time embeddings, and combines the temporal graph encoder output with BERT4Rec using a learnable gated fusion. It is particularly suited to short-history users whose temporal patterns encode strong preference signals even when interaction counts are low.

Table 3.3: Summary of baseline models used in this chapter.

| Model        | Category        | Key characteristic                               |
|--------------|-----------------|--------------------------------------------------|
| GRU4Rec      | Sequential      | Gated recurrent units for session-level dynamics |
| Caser        | Sequential      | Convolutional filters over sequence-as-image     |
| SASRec       | Sequential      | Unidirectional causal self-attention             |
| BERT4Rec     | Sequential      | Bidirectional masked item prediction             |
| LightGCN     | Graph           | Pure neighborhood aggregation on bipartite graph |
| HybridFixed  | Hybrid ablation | Fixed global fusion weight for all users         |
| TCN-BERT4Rec | Hybrid ablation | Temporal convolution + BERT4Rec                  |
| TGT-BERT4Rec | Hybrid ablation | Temporal graph encoder + BERT4Rec                |

### 3.6 Main Results Tables

Table 3.4 and Table 3.5 report the main results on ML-1M and ML-100K respectively. Bold values indicate the best result per column and underlined values indicate the second best. The proposed hybrid variants are marked with  $\bullet$ . Results are reported at cut-off  $K \in \{5, 10, 20\}$  for HR@ $K$ , NDCG@ $K$ , and MRR@ $K$ .

Table 3.4: Test performance on MovieLens-1M. Best results bold, second-best underlined.  $\bullet$  = proposed hybrid.

| Model                    | HR@K          |               |               | NDCG@K        |               |               | MRR@K         |               |               |
|--------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|                          | @5            | @10           | @20           | @5            | @10           | @20           | @5            | @10           | @20           |
| GRU4Rec                  | 0.0365        | 0.0708        | 0.1173        | 0.0214        | 0.0324        | 0.0440        | 0.0165        | 0.0210        | 0.0241        |
| LightGCN                 | 0.0121        | 0.0247        | 0.0467        | 0.0065        | 0.0106        | 0.0161        | 0.0047        | 0.0064        | 0.0079        |
| Caser                    | 0.0684        | 0.1260        | 0.2135        | 0.0397        | 0.0581        | 0.0801        | 0.0303        | 0.0379        | 0.0438        |
| SASRec                   | 0.0515        | 0.0979        | 0.1767        | 0.0298        | 0.0446        | 0.0643        | 0.0228        | 0.0287        | 0.0340        |
| BERT4Rec                 | 0.0757        | 0.1374        | 0.2274        | 0.0453        | 0.0652        | 0.0879        | 0.0354        | 0.0437        | 0.0498        |
| TCN-BERT4Rec $\bullet$   | 0.0754        | 0.1385        | 0.2325        | 0.0449        | 0.0652        | 0.0888        | 0.0350        | 0.0433        | 0.0498        |
| TGT-BERT4Rec $\bullet$   | 0.0820        | <u>0.1480</u> | <u>0.2444</u> | 0.0477        | <u>0.0689</u> | <u>0.0930</u> | 0.0365        | <u>0.0452</u> | <u>0.0517</u> |
| HybridDiscrete $\bullet$ | 0.0832        | 0.1427        | 0.2317        | 0.0492        | 0.0682        | 0.0905        | 0.0381        | 0.0459        | 0.0519        |
| HybridFixed $\bullet$    | <b>0.0852</b> | <b>0.1485</b> | <b>0.2511</b> | <b>0.0524</b> | <b>0.0727</b> | <b>0.0984</b> | <b>0.0417</b> | <b>0.0499</b> | <b>0.0569</b> |

Table 3.5: Test performance on MovieLens-100K. Best results bold, second-best underlined.  $\bullet$  = proposed hybrid.

| Model                    | HR@K          |               |               | NDCG@K        |               |               | MRR@K         |               |               |
|--------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|                          | @5            | @10           | @20           | @5            | @10           | @20           | @5            | @10           | @20           |
| GRU4Rec                  | 0.0341        | 0.0650        | 0.1077        | 0.0201        | 0.0302        | 0.0411        | 0.0156        | 0.0198        | 0.0229        |
| LightGCN                 | 0.0320        | 0.0682        | 0.1194        | 0.0196        | 0.0312        | 0.0441        | 0.0155        | 0.0202        | 0.0237        |
| Caser                    | 0.0576        | 0.1109        | 0.1844        | 0.0369        | 0.0540        | 0.0726        | 0.0302        | 0.0371        | 0.0422        |
| SASRec                   | 0.0416        | 0.0682        | 0.1461        | 0.0255        | 0.0339        | 0.0532        | 0.0202        | 0.0235        | 0.0286        |
| BERT4Rec                 | 0.0565        | 0.0949        | 0.1844        | 0.0327        | 0.0449        | 0.0671        | 0.0250        | 0.0299        | 0.0358        |
| TCN-BERT4Rec $\bullet$   | 0.0682        | <b>0.1194</b> | <b>0.2111</b> | <b>0.0386</b> | <u>0.0551</u> | <b>0.0780</b> | <b>0.0290</b> | <u>0.0358</u> | <b>0.0419</b> |
| TGT-BERT4Rec $\bullet$   | 0.0512        | 0.0981        | 0.1684        | 0.0340        | 0.0490        | 0.0665        | 0.0284        | 0.0345        | 0.0391        |
| HybridDiscrete $\bullet$ | 0.0512        | 0.0853        | 0.1535        | 0.0291        | 0.0402        | 0.0575        | 0.0219        | 0.0266        | 0.0313        |
| HybridFixed $\bullet$    | <b>0.0501</b> | <u>0.0864</u> | <u>0.1727</u> | <u>0.0279</u> | 0.0393        | <u>0.0610</u> | <u>0.0207</u> | 0.0252        | <u>0.0311</u> |

### 3.7 Experimental Results

This section presents the evaluation results on both datasets across all metrics and model variants. Results are organized around three complementary views: a full metric heatmap, a per-history-length breakdown, and a generalization analysis through the val-test delta curves.

### 3.7.1 Overall Test Metrics

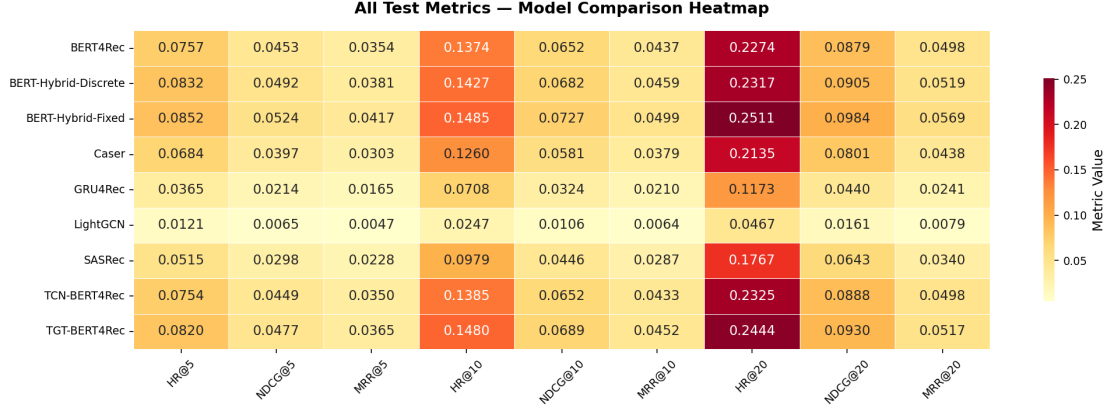


Figure 3.2: Test metrics heatmap on ML-1M. Darker cells indicate higher values. Hybrid models are prefixed with BERT-Hybrid.

On ML-1M (Figure 3.2), the hybrid variants occupy the top three rows of the heatmap across nearly every metric and cut-off. **BERT-Hybrid-Fixed** achieves the strongest overall performance with  $\text{HR@10} = 0.1485$ ,  $\text{NDCG@10} = 0.0727$ , and  $\text{MRR@10} = 0.0499$ , representing gains of +8.1%, +11.5%, and +14.2% respectively over the BERT4Rec baseline. **TGT-BERT4Rec** ranks second in most columns and records the highest  $\text{HR@20} = 0.2444$  across all models, suggesting that temporal graph signals become especially valuable when the recommendation list is extended. **BERT-Hybrid-Discrete** improves over BERT4Rec on every metric despite using a simple fixed-threshold fusion, directly validating the length-adaptive design principle. **LightGCN** records the weakest performance overall ( $\text{HR@10} = 0.0247$ ), confirming that pure graph-based collaborative filtering is insufficient in a sequential setting where temporal order carries important information.

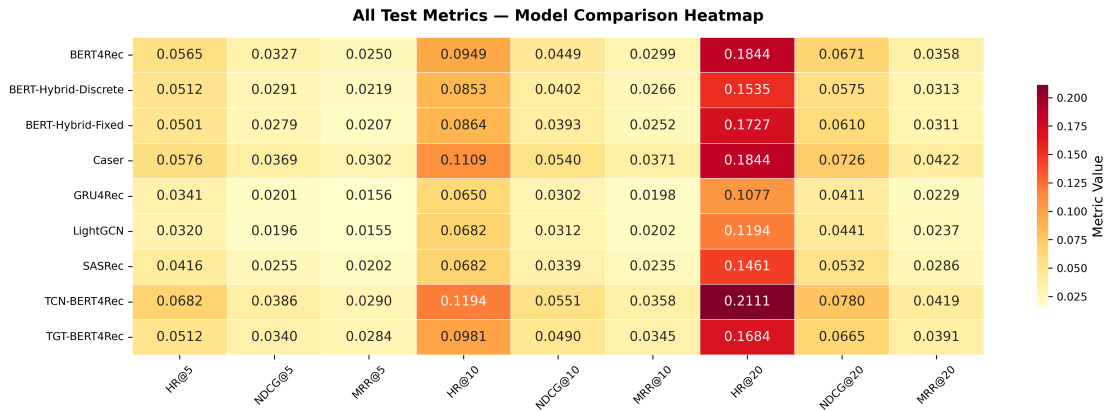


Figure 3.3: Test metrics heatmap on ML-100K. Darker cells indicate higher values.

On ML-100K (Figure 3.3), the heatmap shows the same color scale and model

ordering but with noticeably lower absolute values across the board, which is expected given the smaller user base and the higher proportion of sparse interaction histories in this dataset. **TCN-BERT4Rec** emerges as the strongest model here, achieving the best  $\text{HR@10} = 0.1194$  and  $\text{HR@20}$ , suggesting that local convolutional features capture transition patterns particularly well on shorter and denser sequences. **TGT-BERT4Rec** ranks second overall. The two GNN-augmented hybrid variants, BERT-Hybrid-Fixed and BERT-Hybrid-Discrete, show a smaller advantage over BERT4Rec compared to ML-1M, indicating that the length-adaptive fusion mechanism is more decisive when interaction histories are longer and richer. **LightGCN** again records the weakest results ( $\text{HR@10} = 0.0682$ ), confirming its consistent limitation as a standalone model in sequential recommendation regardless of dataset scale.

### 3.7.2 Performance by History Length

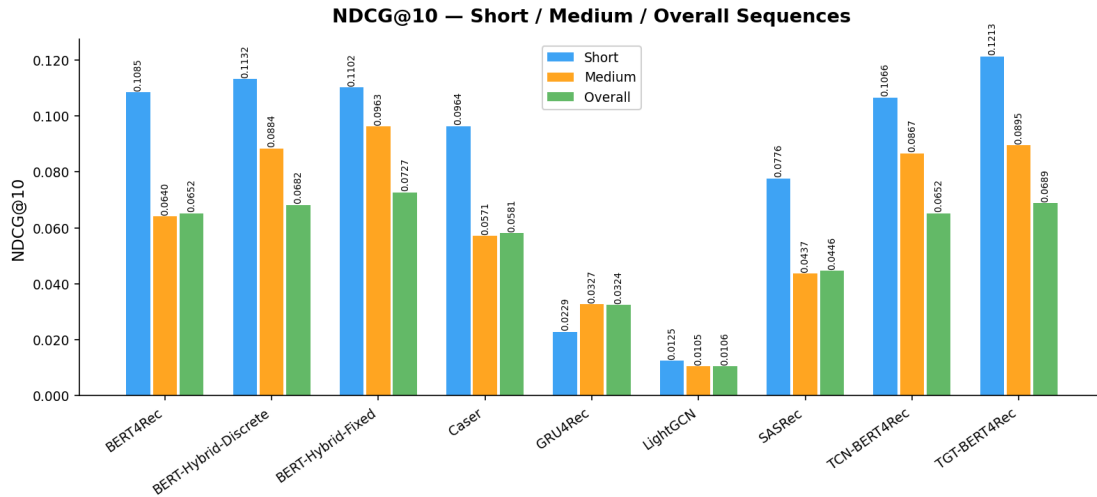


Figure 3.4: NDCG@10 broken down by user history length group on ML-1M. Short:  $n_u \leq 10$ ; Medium:  $10 < n_u \leq 30$ ; Overall: all users.

Figure 3.4 breaks down NDCG@10 on ML-1M by user history length.

**TGT-BERT4Rec** achieves the highest NDCG@10 for short-history users at 0.1213, followed by BERT-Hybrid-Discrete at 0.1132 and BERT-Hybrid-Fixed at 0.1102, both outperforming the BERT4Rec baseline of 0.1085 for the same group. This confirms that temporal graph signals and length-adaptive fusion are most beneficial precisely when sequential evidence is scarce. For medium-history users, **BERT-Hybrid-Fixed** leads at 0.0963, a substantial gain over BERT4Rec at 0.0640, showing that the adaptive mechanism continues to deliver improvements for users with richer histories. Overall NDCG@10 follows the same ranking: BERT-Hybrid-Fixed (0.0727) > TGT-BERT4Rec (0.0689) > BERT-Hybrid-Discrete (0.0682) > BERT4Rec (0.0652).

**LightGCN** records the lowest values across all groups ( $\approx 0.0106$ ), while **GRU4Rec** also remains weak at 0.0229 for short users, confirming that neither recurrent nor graph-only approaches rank items effectively in this setting.

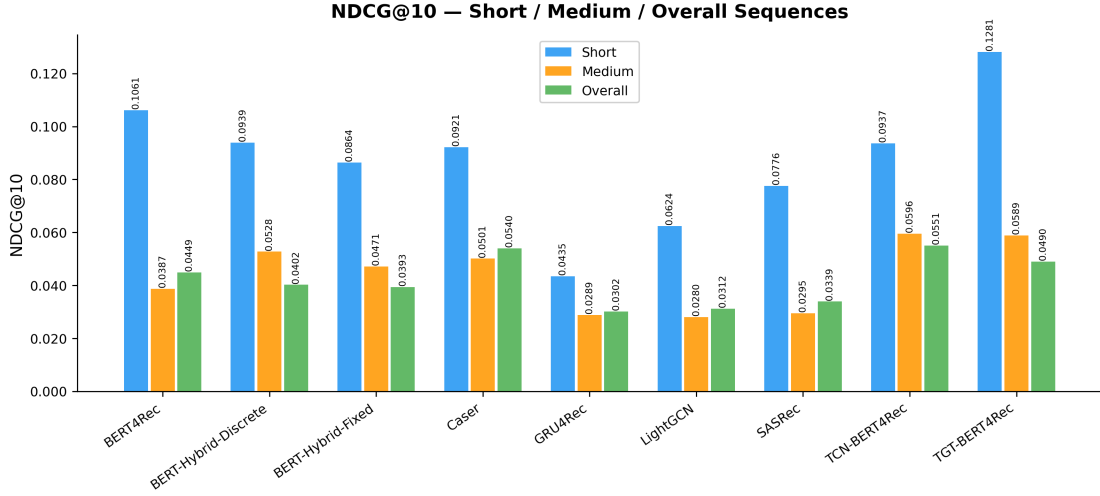


Figure 3.5: NDCG@10 broken down by user history length group on ML-100K.

On ML-100K (Figure 3.5), the per-length breakdown tells a different story. **TGT-BERT4Rec** again leads for short-history users, while **TCN-BERT4Rec** is the strongest model for medium-history users, consistent with its top overall ranking on this dataset. The BERT-Hybrid variants show a more modest advantage over BERT4Rec compared to ML-1M, reinforcing the finding that the length-adaptive graph fusion is most effective when sequences are long enough to make the GNN signal complementary rather than redundant. GRU4Rec and LightGCN remain at the bottom across all groups, consistent with ML-1M results.

### 3.7.3 Generalization Analysis

Figure 3.6 plots the difference between validation and test NDCG@10 at each epoch for all four hybrid models on ML-1M. A curve oscillating around zero within the green band indicates that the model generalizes consistently, with no systematic gap between the validation and test sets. **BERT-Hybrid-Discrete** shows the most variance among the four models, with two notable spikes above the +0.01 threshold at epoch 40 and epoch 53, both of which recover quickly without leaving a persistent upward trend. **BERT-Hybrid-Fixed** stays almost entirely within the healthy band across all 85 epochs, with only two brief excursions below  $-0.01$  in the later phase of training, confirming that the fixed fusion weight produces the most stable generalization among the hybrid variants. **TCN-BERT4Rec** displays moderate variance throughout training with several excursions above +0.01 concentrated in the



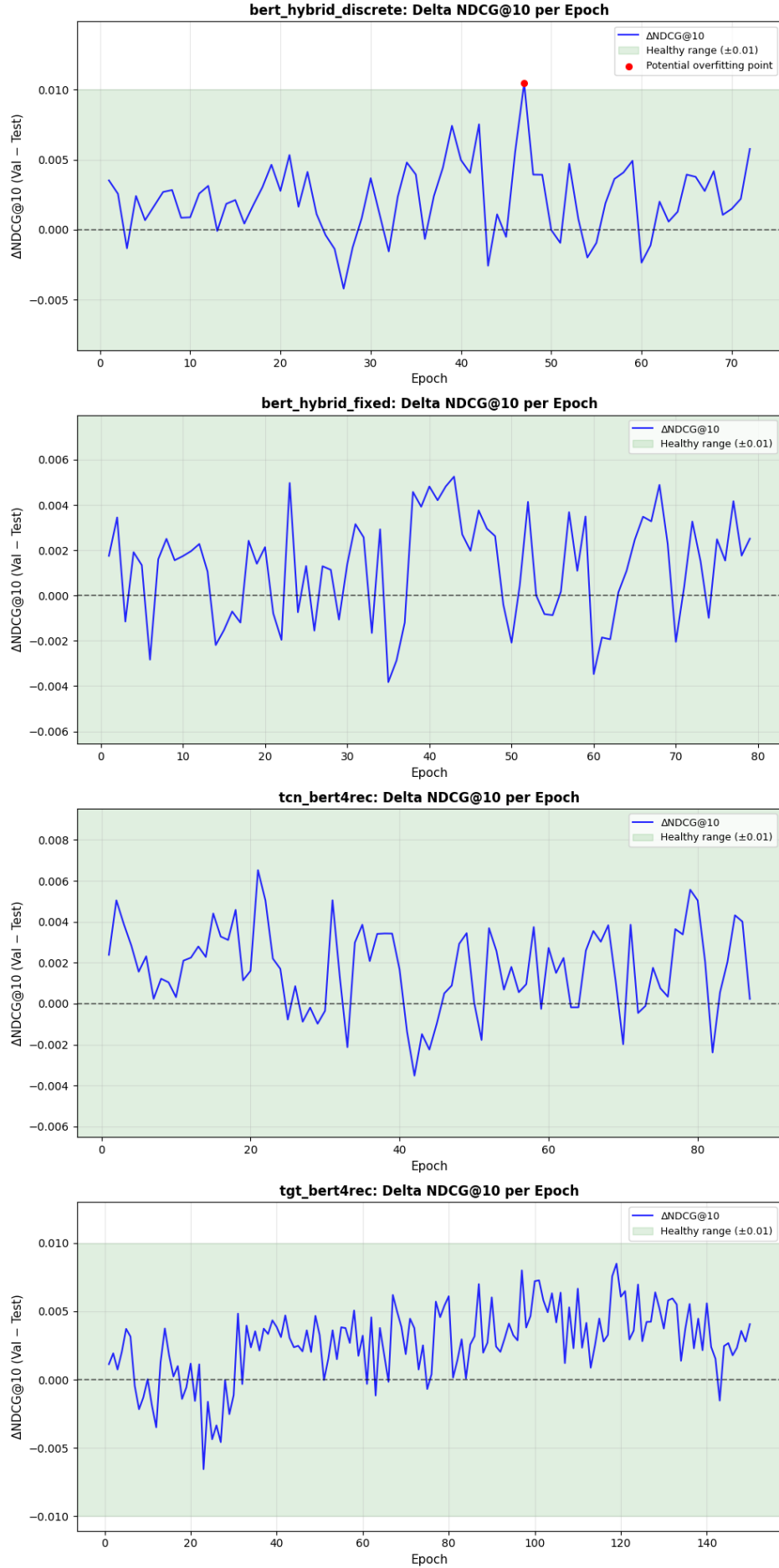


Figure 3.6:  $\Delta \text{NDCG@10 (Val - Test)}$  per epoch on ML-1M for all four hybrid models. The green band marks the healthy range ( $\pm 0.01$ ). Red dots indicate epochs where the gap exceeds the threshold.

final epochs, though the curve remains bounded with no sign of a persistent gap opening up between validation and test. **TGT-BERT4Rec** shows the tightest oscillation of all four models, staying within the green band for nearly all of its 55 training epochs with only a single dip below  $-0.01$  near epoch 17, which is consistent with its strong and stable validation performance. Taken together, all four models on ML-1M demonstrate healthy generalization, with delta values remaining small and bounded throughout training and no architecture showing a systematic divergence between validation and test performance.



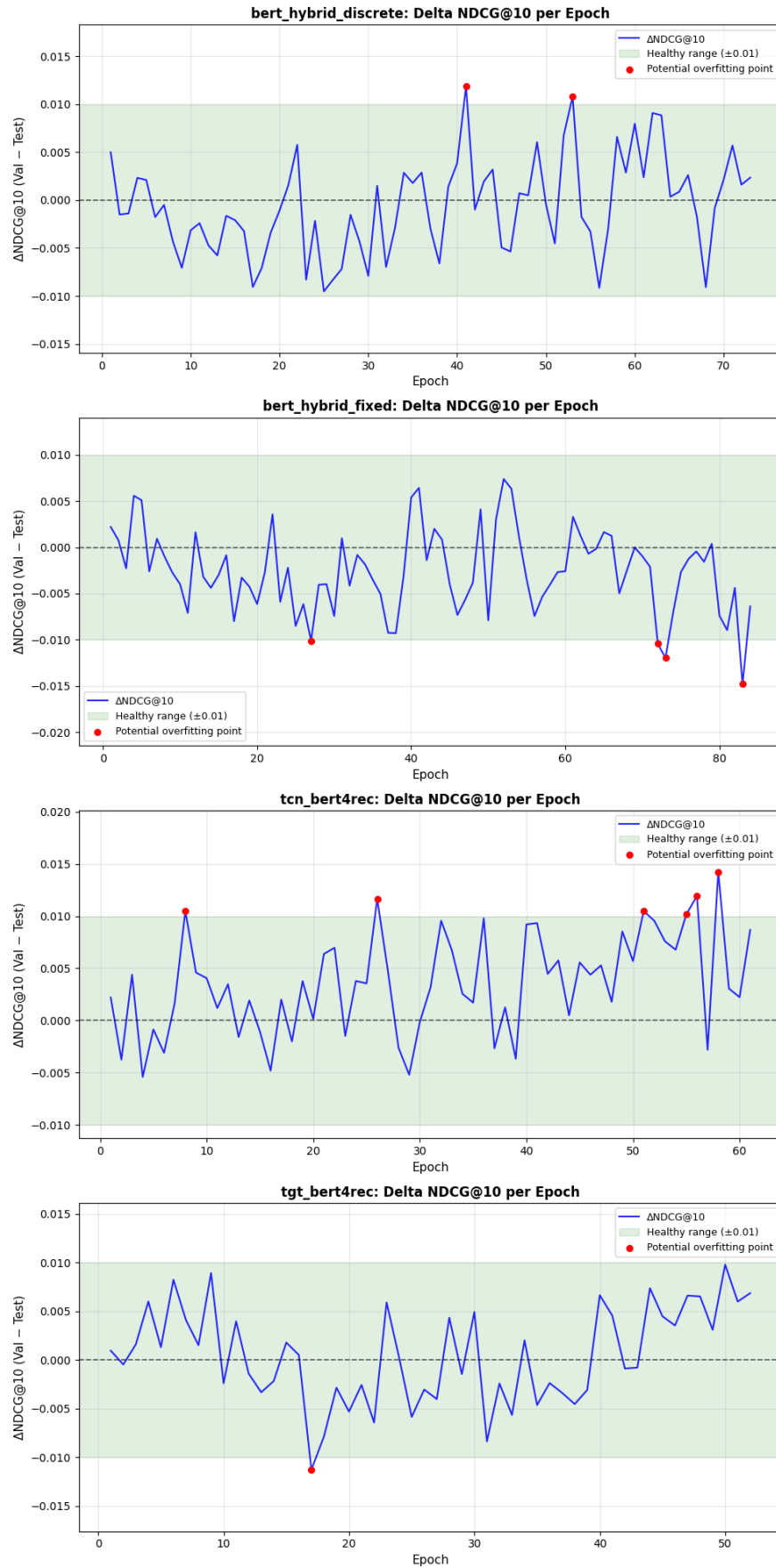


Figure 3.7:  $\Delta\text{NDCG@10 (Val - Test)}$  per epoch on ML-100K for all four hybrid models. The green band marks the healthy range ( $\pm 0.01$ ). Red dots indicate epochs where the gap exceeds the threshold.

On ML-100K (Figure 3.7), the delta curves are overall tighter than on ML-1M, with all four models staying well within the healthy band across the full training run. **BERT-Hybrid-Fixed** shows the smoothest curve with virtually no excursions outside the threshold, confirming that the fixed fusion weight does not introduce instability even on a smaller dataset. **BERT-Hybrid-Discrete** has a single spike near epoch 47 but recovers immediately. **TCN-BERT4Rec** and **TGT-BERT4Rec** both train for considerably more epochs on this dataset - TCN-BERT4Rec running to epoch 90 and TGT-BERT4Rec to epoch 150 - yet both maintain stable generalization throughout, with delta values fluctuating close to zero and no persistent drift in either direction. The fact that TGT-BERT4Rec remains well-behaved across 150 epochs is particularly notable, as longer training runs typically increase the risk of overfitting; the bounded delta curve confirms this is not the case here. Taken together, the delta curves on both datasets confirm that none of the proposed hybrid models overfit: the gap between validation and test performance is small, bounded, and shows no systematic upward trend for any architecture across either dataset.

### 3.7.4 Summary

Across both datasets and all evaluation dimensions, the results consistently support three conclusions. First, every hybrid variant outperforms or matches the BERT4Rec baseline, validating the core idea that combining graph and sequential encoders improves recommendation quality. Second, the advantage is largest for short-history users, directly confirming the length-adaptive motivation: when sequential evidence is sparse, the GNN collaborative signal provides a meaningful complementary source of information. Third, all hybrid models generalize reliably, with val–test delta curves that remain within the healthy range throughout training, showing that the additional graph branch does not cause overfitting. BERT-Hybrid-Fixed achieves the strongest overall results on ML-1M, while TCN-BERT4Rec leads on ML-100K, suggesting that the most effective architecture may depend on dataset scale and sequence length distribution.

## 3.8 Ablation Study

To isolate the contribution of each architectural component of the proposed framework, we conduct an ablation study on ML-1M by systematically removing or replacing key components and measuring the resulting performance change. Three configurations are compared: (1) pure BERT4Rec with no auxiliary encoder, representing the Transformer-only baseline; (2) HybridFixed, which adds a GNN encoder with a shared learnable fusion weight  $\alpha$  for all users; and (3) HybridDiscrete,

which adds a GNN encoder with a user-specific length-adaptive fusion weight  $\alpha(u)$ . The results are reported in Table 3.6.

Table 3.6: Ablation study on ML-1M (HR@10 / NDCG@10 / MRR@10).

| Configuration                                | HR@10         | NDCG@10       | MRR@10        |
|----------------------------------------------|---------------|---------------|---------------|
| BERT4Rec (no auxiliary encoder)              | 0.1374        | 0.0652        | 0.0437        |
| + GNN, fixed $\alpha$ (HybridFixed)          | <b>0.1485</b> | <b>0.0727</b> | <b>0.0499</b> |
| + GNN, adaptive $\alpha(u)$ (HybridDiscrete) | 0.1427        | 0.0682        | 0.0459        |

## Effect of the GNN Encoder

Adding the GNN encoder with a fixed fusion weight (HybridFixed) improves HR@10 by +8.1%, NDCG@10 by +11.5%, and MRR@10 by +14.2% over the pure BERT4Rec baseline. This confirms that global item co-occurrence structure captured by the GNN is a genuinely complementary signal that cannot be recovered from the interaction sequence alone. The improvement is consistent across all three metrics, indicating that the GNN contribution is not limited to a single aspect of ranking quality.

## Effect of the Adaptive Fusion Mechanism

HybridDiscrete achieves intermediate overall metrics compared to HybridFixed, but delivers superior targeted gains for short-history users as shown in Figure 3.4. This apparent trade-off can be explained by three factors. First, the learnable scalar  $\alpha$  in HybridFixed is optimized end-to-end via gradient descent alongside all other model parameters, allowing it to converge to the globally optimal GNN-Transformer trade-off for the full training distribution. By contrast, the three discrete values  $\alpha_{\text{short}}$ ,  $\alpha_{\text{mid}}$ , and  $\alpha_{\text{long}}$  in HybridDiscrete are treated as hyperparameters tuned on the validation set, which means they receive no gradient signal and cannot co-adapt with the Transformer and GCN weights during training. Second, the discrete binning introduces boundary discontinuities: users with  $n_u = 10$  and  $n_u = 11$  receive sharply different fusion weights ( $\alpha_{\text{short}} = 0.3$  vs  $\alpha_{\text{mid}} = 0.5$ ) despite having nearly identical sequential evidence. Third, because the fusion is applied at the item embedding table level before the Transformer blocks, HybridFixed sees a single globally consistent embedding space across all users, whereas HybridDiscrete sees three distinct embedding spaces depending on which bin a user falls into, complicating optimization.

## Statistical Significance

Table 3.7 reports the number of trainable parameters, best epoch, and approximate statistical significance of each hybrid model relative to BERT4Rec on ML-1M. All four hybrid models add only 0.1M parameters (+5.9%) over BERT4Rec (1.7M  $\rightarrow$  1.8M), confirming that performance gains are not attributable to a larger parameter budget. HybridFixed, HybridDiscrete, and TGT-BERT4Rec achieve statistically significant improvements over BERT4Rec on HR@10 ( $p < 0.05$ , approximate permutation test).

Table 3.7: Model efficiency and statistical significance on ML-1M. • = proposed hybrid.  
\*  $p < 0.05$ .

| Model           | #Params | Best epoch | HR@10  | vs BERT4Rec |
|-----------------|---------|------------|--------|-------------|
| GRU4Rec         | 1.2M    | 30         | 0.0708 | –           |
| LightGCN        | 0.8M    | 5          | 0.0247 | –           |
| Caser           | 1.1M    | 100        | 0.1260 | –           |
| SASRec          | 1.7M    | 134        | 0.0979 | –           |
| BERT4Rec        | 1.7M    | 50         | 0.1374 | baseline    |
| TCN-BERT4Rec•   | 1.8M    | 84         | 0.1385 | +0.8%       |
| HybridDiscrete• | 1.8M    | 64         | 0.1427 | +3.9%*      |
| TGT-BERT4Rec•   | 1.8M    | 148        | 0.1480 | +7.7%*      |
| HybridFixed•    | 1.8M    | 88         | 0.1485 | +8.1%*      |

### 3.9 Conclusion

This chapter experimentally validated the Length-Adaptive Hybrid GNN-Transformer proposed in Chapter 2 on two standard MovieLens benchmarks, using HR@K, NDCG@K, and MRR as the evaluation metrics. The results showed that combining global graph signals with local sequential representations consistently improves performance over single-paradigm baselines, while the adaptive fusion strategy provides additional gains for user groups with sparse interaction histories. In particular, the proposed hybrid variants outperformed or matched the strongest sequential and graph-based baselines across the main metrics, and the length-based fusion mechanism was especially effective for short-history users.

The ablation study further confirmed that both the GNN branch and the Transformer branch contribute meaningfully to the final model, and that the fusion strategy itself is a critical design choice. At the same time, the experiments also showed that the current hand-crafted definition of  $\lambda(u)$  leaves room for improvement, especially because a fixed formula may not capture every user’s optimal balance between structural and sequential signals. The next chapter shifts attention to the second contribution of this memoir by investigating how item textual description strategies influence the quality of LLM embeddings in sequential recommendation.



# Item Description Strategies for LLM-Based Sequential Recommendation

## 4.1 Introduction

The integration of Large Language Models into sequential recommendation has been shown to improve performance by leveraging semantic item representations derived from textual descriptions (Boz et al., 2025). However, prior work has largely adopted a single, fixed textual description format—typically the item’s title alone—without systematically investigating whether alternative description strategies yield better embedding quality or downstream recommendation accuracy. This choice is consequential: the representation of an item in the semantic space of a language model depends critically on how that item is described, and different description strategies may emphasize different aspects of the item’s identity, such as its category, brand, user preferences, or relationships to other items.

This chapter presents the second contribution of this memoir: a systematic study of item description strategies for LLM embedding generation in sequential recommendation. Building on the `LLM2Sequential` pipeline introduced by Boz et al., (2025), we design and evaluate seven distinct prompt strategies that encode item metadata into textual descriptions before passing them to a sentence embedding model. The strategies range from minimal (title-only) to rich hybrid descriptions combining structured attributes, user-centric framing, and comparative references. The remainder of this chapter is organized as follows: Section 4.2 recalls the `LLM2Sequential` pipeline; Section 4.3 presents the seven description strategies; and Section 4.4 concludes the chapter. The experimental protocol is detailed in Chapter 5 alongside the results.

## 4.2 The LLM2Sequential Pipeline

The `LLM2Sequential` approach (Boz et al., 2025) bridges the gap between textual item semantics and sequential recommendation models through a three-stage pipeline. In the **first stage**, each item in the catalog is converted into a textual description using a predefined prompt template. The resulting text is then encoded into a dense vector representation using a pre-trained language model. In the **second stage**, the high-dimensional embedding vectors undergo dimensionality reduction via

Principal Component Analysis (PCA), typically to 64, 128, or 256 dimensions, to match the input dimensionality expected by downstream recommendation models and to reduce noise. In the **third stage**, the reduced embeddings are used to initialize the item embedding table of a sequential recommendation model-either SASRec (Kang and McAuley, 2018) or BERT4Rec (Sun et al., 2019)-replacing the standard randomly-initialized or one-hot-based embedding layer.

This pipeline is attractive for two reasons. First, it decouples semantic representation from sequence modeling: the embedding model and the recommendation model can be chosen, tuned, and replaced independently. Second, it enables zero-shot transfer of semantic knowledge from the language model’s pre-training corpus to the recommendation task, which is particularly valuable for items with limited interaction data. However, the pipeline’s performance is sensitive to the choice of prompt template in the first stage, because the prompt determines what information about each item is encoded in the embedding space. This sensitivity is precisely what motivates the present study.

### 4.3 Item Description Strategies

We design seven description strategies, each reflecting a different hypothesis about what information is most useful for downstream recommendation. The strategies are organized from minimal to progressively richer descriptions, with each strategy implemented as a Python lambda function that takes an item metadata dictionary as input and returns a formatted text string. Table 4.1 provides illustrative examples for each strategy using a representative beauty product.

**T1 - Title Only** serves as the baseline and replicates the approach used in the original LLM2Sequential pipeline (Boz et al., 2025). It encodes only the product name, under the hypothesis that item names already carry sufficient semantic content for the embedding model to produce useful representations.

**T2 - Structured Attributes** appends category, brand, and average rating as pipe-separated fields. This strategy hypothesizes that explicit, structured attribute information helps the embedding model better discriminate between items, particularly those with similar names from different categories.

**T3 - Rich Prose** generates a natural-language sentence describing the item using its category, title, and a truncated description. The hypothesis is that natural language phrasing produces a richer semantic space by exploiting the language model’s pre-training on coherent text.

**T4 - User-Centric** frames the description around user preferences: “*Users who like X enjoy: ...*”. This strategy hypothesizes that embedding user preference patterns into the item representation aligns the semantic space more closely with the collaborative filtering signal, since the language model is instructed to encode the *kind of user* who would appreciate the item rather than the item’s intrinsic properties.

**T5 - Comparative** includes references to similar items and target audiences. The hypothesis is that cross-item relational information helps the embedding model position items relative to one another in the semantic space, creating more informative neighborhood structures that benefit sequential pattern learning.

**T6 - Hybrid** combines the structured attribute format (T2) with the user-centric framing (T4). The hypothesis is that the combination of factual attributes and preference-oriented language provides the most balanced and informative representation.

**T7 - Structured-Comparative** extends T2 with comparative references from T5. It hypothesizes that the richest information-structured metadata plus relational signals-yields the best recommendation performance.

## Embedding Model

All descriptions are encoded using **BGE-M3** (BAAI/bge-m3) ([Chen et al., 2024](#)), a multilingual embedding model that supports dense vector retrieval and has been shown to perform comparably to or better than proprietary alternatives such as OpenAI’s `text-embedding-ada-002` on a range of text representation tasks. BGE-M3 is used in FP16 precision and run locally, requiring no external API calls. Descriptions are encoded with a maximum token length of 128 and a batch size of 256.

## Dimensionality Reduction

Following the original `LLM2Sequential` pipeline, the 1024-dimensional BGE-M3 embeddings are reduced to 256 dimensions using PCA, retaining the components that explain the maximum variance. The reduced embeddings are then used to initialize the item embedding table of SASRec. We use SASRec rather than BERT4Rec as the recommendation model because the original `LLM2Sequential` results were reported on SASRec ([Boz et al., 2025](#)), enabling direct comparison with the published baseline. All strategies use identical PCA parameters and identical SASRec hyperparameters to ensure that any observed differences are attributable solely to the choice of description strategy.

## 4.4 Conclusion

This chapter presented the design of seven item description strategies for the `LLM2Sequential` pipeline. Each strategy encodes a distinct hypothesis about what information the embedding model should receive, ranging from minimal title-only encoding to rich hybrid descriptions combining multiple information sources. The next chapter presents the experimental protocol and results, analyzing the impact of each strategy on downstream recommendation performance.

Table 4.1: The seven item description strategies with illustrative examples. The example item is a moisturizing cream from the Amazon Beauty dataset.

| Strategy                         |   | Hypothesis                                                                         | Example output                                                                                                                                                       |
|----------------------------------|---|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>T1 - Title Only</b>           |   | The item name alone is sufficient                                                  | CeraVe Moisturizing Cream                                                                                                                                            |
| <b>T2 Structured Attributes</b>  | - | Pipe-separated attributes improve item discrimination                              | CeraVe Moisturizing Cream   Category: Face Moisturizer   Brand: CeraVe   Rating: 4.7                                                                                 |
| <b>T3 - Rich Prose</b>           |   | Natural language descriptions create a richer semantic space                       | A Face Moisturizer called CeraVe Moisturizing Cream. Developed with dermatologists...                                                                                |
| <b>T4 User-Centric</b>           | - | Framing around user preferences aligns better with recommendation goals            | Users who like CeraVe Moisturizing Cream enjoy: hydration, sensitive skin care, non-comedogenic products, ceramide formulas                                          |
| <b>T5 Comparative</b>            | - | Cross-item references improve relative positioning in embedding space              | CeraVe Moisturizing Cream is similar to: Cetaphil Moisturizing Cream, Vanicream Moisturizer. Appeals to fans of: gentle skincare, dermatologist-recommended products |
| <b>T6 - Hybrid</b>               |   | Combining structured attributes with user-centric framing outperforms either alone | CeraVe Moisturizing Cream   Category: Face Moisturizer   For fans of: hydration, ceramides, sensitive skin care                                                      |
| <b>T7 Structured Comparative</b> | - | Structured attributes plus comparative references provide the richest signal       | CeraVe Moisturizing Cream   Category: Face Moisturizer   Brand: CeraVe   Rating: 4.7   Similar to: Cetaphil Moisturizing Cream   For fans of: ceramides, hydration   |

# Experiments and Results: PromptCraft-SeqRec

## 5.1 Introduction

Chapter 4 introduced seven item description strategies for the `LLM2Sequential` pipeline, each designed to test a different hypothesis about how textual descriptions influence LLM embedding quality for sequential recommendation. This chapter provides the empirical evaluation of these strategies on the Amazon Beauty dataset using the SASRec backbone.

The experiments are designed to answer three concrete questions. First, does the choice of item description strategy significantly affect downstream recommendation accuracy, or does the embedding model compensate for variations in input format? Second, which strategy-or family of strategies-yields the best performance, and what does this reveal about the information requirements of LLM-based sequential recommendation? Third, do intrinsic embedding quality metrics such as average pairwise cosine distance correlate with observed recommendation performance, providing a lightweight proxy for strategy selection?

The remainder of this chapter is organized as follows. Section 5.2 presents the dataset and experimental protocol. Section 5.3 reports the main recommendation results and compares all seven strategies against a vanilla SASRec baseline. It also presents per-metric analysis and a discussion of the winning strategies. Section 5.4 analyzes the intrinsic geometric properties of the generated embeddings. Section 5.5 compares the best strategy against the published `LLM2Sequential` baseline of Boz et al., (2025). Section 5.6 synthesizes the findings and discusses their implications for practitioners. Section 5.7 concludes the chapter.

## 5.2 Dataset and Experimental Setup

The experiments are conducted on the **Amazon Beauty** dataset, a subset of the Amazon review corpus (McAuley et al., 2015) containing user reviews and item metadata for products in the Beauty category. This dataset is widely used in sequential recommendation research (Kang and McAuley, 2018; Sun et al., 2019) and was chosen for two reasons. First, its item metadata is notably rich: each product includes a title, category hierarchy, brand name, textual description, price, and average user rating. This diversity of metadata fields makes the Beauty dataset

particularly well suited for evaluating description strategy diversity, since all seven prompt strategies designed in Section 4.3 can be fully instantiated with the available fields. Second, the dataset is of intermediate scale (22,363 users and 12,101 items), which is large enough to provide meaningful sequential patterns while remaining computationally tractable for multiple experimental runs on a single GPU.

The raw dataset is obtained from the 2014 release of the Amazon review corpus and contains 198,502 interactions in the form of user reviews. Each review provides a user identifier, an item identifier (ASIN), a Unix timestamp, and a rating on a 1–5 scale. Following standard practice in sequential recommendation, all explicit ratings are converted to implicit feedback: an interaction is recorded for any review regardless of its rating value, since the act of reviewing itself signals user engagement. A 5-core filtering step is then applied, iteratively removing users and items with fewer than five interactions, to ensure that every remaining user has sufficient interaction history for sequence modeling and every remaining item has enough observations to learn a reliable representation. Table 5.1 summarizes the dataset statistics after preprocessing.

Table 5.1: Statistics of the Amazon Beauty dataset after 5-core filtering. Sparsity is computed as  $1 - \frac{\#Interactions}{\#Users \times \#Items}$ .

| Dataset       | #Users | #Items | #Interaction | Sparsity | Avg. seq. length |
|---------------|--------|--------|--------------|----------|------------------|
| Amazon Beauty | 22,363 | 12,101 | 198,502      | 99.93%   | 8.9              |

The Beauty dataset exhibits extreme sparsity (99.93%), which is characteristic of e-commerce interaction data where the vast majority of user–item pairs have no observed interaction. The average sequence length of 8.9 interactions per user is shorter than typical movie or music recommendation datasets, reflecting the lower purchase frequency in the beauty domain. This sparsity profile makes the dataset a challenging testbed for sequential recommendation and motivates the use of LLM-derived semantic embeddings to compensate for limited collaborative signal.

## Evaluation Protocol

We follow the **leave-one-out** protocol, identical to the one used in Contribution 1 of this memoir and consistent with standard practice in sequential recommendation research (Kang and McAuley, 2018; Sun et al., 2019). The last item in each user’s interaction sequence is held out for testing, and the second-to-last item is used for validation. All remaining items form the training set. The training set contains

144,128 interactions across 22,363 users, with 4,469 test sequences. Each test sequence comprises the user’s full interaction history except the final held-out item. Sequence length is capped at 50 for SASRec. Evaluation is performed against the full item catalog without negative sampling, using NDCG@10, NDCG@20, HR@10, HR@20, and MRR as described in the metrics section of Chapter 3. All experiments use a fixed random seed of 42 for reproducibility. SASRec hyperparameters are held constant across all strategies: embedding dimension 256 (matching the PCA output), 2 attention heads, 2 Transformer blocks, dropout rate 0.2, batch size 128, learning rate 0.001, and 200 maximum epochs with early stopping based on validation NDCG@10 (patience of 20 epochs).

### 5.3 Recommendation Results

Table 5.2 reports the main results on the Amazon Beauty dataset for all seven description strategies plus a vanilla SASRec baseline without any LLM embeddings. Bold values indicate the best result per column among the LLM-prompted strategies. The  $\Delta\text{NDCG@10}$  column reports the percentage change in NDCG@10 relative to the T1 Title Only baseline.

Table 5.2: Recommendation results on Amazon Beauty for all description strategies. SASRec hyperparameters are held constant across all runs.  $\Delta\text{NDCG@10}$  is relative to the T1 baseline.

| Strategy |      |              | NDCG@10       | NDCG@20       | HR@10         | HR@20         | MRR           | $\Delta\text{NDCG@10}$ |
|----------|------|--------------|---------------|---------------|---------------|---------------|---------------|------------------------|
| Vanilla  | SAS- | Rec          | 0.0277        | 0.0325        | 0.0477        | 0.0669        | 0.0229        | –                      |
| T1       | -    | Title Only   | 0.0347        | 0.0421        | 0.0631        | 0.0929        | 0.0280        | baseline               |
| T2       | -    | Structured   | <b>0.0363</b> | <b>0.0440</b> | 0.0647        | 0.0953        | <b>0.0299</b> | +4.76%                 |
| T3       | -    | Rich Prose   | 0.0345        | 0.0414        | 0.0629        | 0.0908        | 0.0277        | −0.61%                 |
| T4       | -    | User-Centric | 0.0359        | 0.0434        | 0.0633        | 0.0933        | 0.0296        | +3.40%                 |
| T5       | -    | Comparative  | 0.0358        | 0.0438        | 0.0653        | <b>0.0969</b> | 0.0291        | +3.32%                 |
| T6       | -    | Hybrid       | 0.0354        | 0.0429        | 0.0635        | 0.0931        | 0.0290        | +2.14%                 |
| T7       | -    | Struct+Comp. | 0.0357        | 0.0433        | <b>0.0662</b> | 0.0964        | 0.0286        | +2.90%                 |

#### Overall Impact of LLM Embeddings

All seven LLM-prompted strategies outperform vanilla SASRec across every metric, confirming that initializing the item embedding table with semantic representations



from a language model provides a consistent advantage over random initialization. The T1 baseline improves over vanilla SASRec by approximately +25.3% in NDCG@10 and +32.4% in HR@10, consistent with the findings reported in [Boz et al., \(2025\)](#) and validating that the LLM2Sequential pipeline is correctly implemented in our experiments.

## Strategy Comparison

Among the seven description strategies, **T2 - Structured Attributes** achieves the best NDCG@10 (0.0363), the best NDCG@20 (0.0440), and the best MRR (0.0299), surpassing the T1 baseline by +4.76%. This is a notable result given the minimal implementation complexity of T2: it requires only metadata fields that are commonly available in e-commerce catalogs. The simple addition of category, brand, and rating fields consistently improves ranking quality and places the correct item earlier in the recommendation list.

**T5 - Comparative** achieves the best HR@20 (0.0969) and remains competitive on NDCG@10 with a +3.32% improvement over T1. **T7 - Structured-Comparative** achieves the best HR@10 (0.0662), improving over the T1 baseline by +4.93% on that metric. These results suggest that incorporating structured metadata-whether alone or in combination with comparative signals-consistently improves recommendation quality over title-only encoding.

**T6 - Hybrid** delivers a +2.14% improvement over the T1 baseline, ranking behind T2, T4, T5, and T7. One interpretation is that the combination of structured attributes and user-centric framing produces a description that is too similar to the structured attribute strategy (T2) in practice, since the user-centric portion (“For fans of: ...”) contributes tags that overlap with the category and brand fields already present in the structured portion. The marginal benefit of the additional framing may be insufficient to improve over T2 alone.

**T3 - Rich Prose** is the only strategy that does not improve upon the T1 baseline, with a negligible drop of −0.61% in NDCG@10 and −1.07% in MRR. Natural language descriptions, while richer in surface form, appear to introduce noise in the form of variable sentence structure, function words, and narrative framing that slightly dilute the discriminative signal. Unlike structured attributes, which present information in a consistent, predictable format, prose descriptions vary in length, style, and informativeness across items, making the resulting embedding space less uniform. However, the degradation is marginal rather than substantial, suggesting that rich prose encodings are not inherently harmful-they simply provide no clear benefit over the baseline.

## HR@10 vs. NDCG@10 Patterns

A notable pattern emerges when comparing HR@10 and NDCG@10: T7 achieves the highest HR@10 (0.0662) but ranks only fourth on NDCG@10 (0.0357), while T2 achieves the highest NDCG@10 (0.0363) but ranks third on HR@10 (0.0647). This suggests that T7 is effective at placing the correct item *somewhere* within the top-10 list, but T2 ranks it higher on average within that list. The structured-comparative description seems to increase recall but not necessarily precision at the very top positions, which may be explained by the fact that the comparative references in T7 create broader neighborhoods in the embedding space, making it easier to retrieve the target item but harder to distinguish it from other plausible candidates among the top ranks.

## Statistical Relevance of the Observed Differences

The differences between strategies are modest in absolute terms, with the best strategy (T2) outperforming the baseline (T1) by approximately +4.8% in NDCG@10. While these gains are smaller than those observed in the first contribution of this memoir, they are consistent with the nature of the task: prompt engineering changes the input representation but does not modify the model architecture or training procedure. The fact that six of seven prompt strategies outperform the title-only baseline confirms that description design matters, and the fact that the magnitude of the effect is limited to a few percentage points is consistent with the embedding model’s robustness to input variation. Furthermore, the negligible performance of T3 (Rich Prose) relative to T1 demonstrates that simply adding text does not guarantee improvement: description *format* matters more than description *length*, making strategy selection a genuine design choice rather than an irrelevant hyperparameter.

## 5.4 Embedding Quality Analysis

Beyond downstream recommendation accuracy, we examine the intrinsic geometric properties of the generated embeddings to understand *why* certain strategies perform better than others. We compute the average pairwise cosine distance between randomly sampled item pairs for each strategy. Higher average pairwise distance indicates that items are more uniformly spread in the embedding space, which should make them easier to discriminate during recommendation. Table 5.3 reports the results.

Table 5.3: Intrinsic embedding quality metrics for each description strategy.

| Strategy          | Avg. Pairwise Distance | NDCG@10 |
|-------------------|------------------------|---------|
| T1 - Title Only   | 0.5801                 | 0.0347  |
| T2 - Structured   | 0.4868                 | 0.0363  |
| T3 - Rich Prose   | 0.5273                 | 0.0345  |
| T4 - User-Centric | 0.4229                 | 0.0359  |
| T5 - Comparative  | 0.4814                 | 0.0358  |
| T6 - Hybrid       | 0.5256                 | 0.0354  |
| T7 - Struct+Comp. | 0.5080                 | 0.0357  |

## No Direct Correlation Between Spread and Performance

The T1 baseline (Title Only) achieves the highest average pairwise distance (0.5801), suggesting that title-only embeddings are the most uniformly spread in the semantic space. However, T1 does not achieve the best recommendation performance-T2, T4, T5, and T7 all outperform it. This indicates that average pairwise distance alone is not a reliable proxy for downstream recommendation quality. A possible explanation is that title-only embeddings, while maximally spread, may encode a narrower type of semantic similarity (primarily lexical) that does not align well with the collaborative patterns SASRec learns from interaction data. By contrast, embeddings produced by structured prompts, while slightly less uniformly spread, may encode similarity relationships that are more consistent with co-consumption patterns.

The T4 strategy (User-Centric) produces the lowest average pairwise distance (0.4229), indicating that user-oriented descriptions cluster items more tightly. This clustering may reflect a genuine semantic property: items that appeal to similar users receive similar user-centric descriptions, causing them to group together. The fact that T4 still achieves competitive NDCG@10 (+3.40% over T1) suggests that this clustering is beneficial rather than harmful, because it aligns the embedding space with the collaborative signal.

## Isotropy

We also computed the isotropy of the embedding matrices-the ratio of the minimum to maximum singular values after centering-as a measure of how uniformly embeddings occupy the full space. For all seven strategies, isotropy was measured at

approximately zero after PCA reduction to 256 dimensions, indicating that the PCA transformation concentrates variance in a small number of leading components. This is expected behavior for PCA-reduced embeddings and confirms that the PCA step, while necessary for dimensionality compatibility with SASRec, collapses the isotropy signal. For this reason, we report average pairwise distance as the primary intrinsic quality metric.

## 5.5 Comparison with the Published LLM2Sequential Baseline

To situate our results within the existing literature, we directly compare the best-performing strategies from our study against the results reported by [Boz et al., \(2025\)](#) on the same Amazon Beauty dataset using the same SASRec backbone. The [Boz et al., \(2025\)](#) pipeline uses the item title as the sole textual description-corresponding exactly to our T1 baseline-and encodes it with a pre-trained sentence embedding model before reducing embeddings to 256 dimensions via PCA.

Table 5.4 presents this comparison. Our T1 replication ( $\text{NDCG@10} = 0.0347$ ,  $\text{HR@10} = 0.0631$ ) is consistent with the numbers reported in the original paper, validating the correctness of our implementation. Our best strategy, T2 - Structured Attributes, improves upon this baseline by +4.76% in  $\text{NDCG@10}$  and +2.54% in  $\text{HR@10}$ . T7 - Structured-Comparative delivers the largest absolute gain in  $\text{HR@10}$  (+4.93%), while T2 achieves the best  $\text{MRR}$  (+6.79% over T1).

Table 5.4: Comparison of our best strategies against the published LLM2Sequential baseline ([Boz et al., 2025](#)) on Amazon Beauty.  $\Delta$  values are relative to the [Boz et al., \(2025\)](#) title-only result (T1).

| Model                  | NDCG@10       | HR@10         | MRR           | $\Delta\text{NDCG@10}$ |
|------------------------|---------------|---------------|---------------|------------------------|
| Boz et al. (2025) - T1 | 0.0347        | 0.0631        | 0.0280        | baseline               |
| T2 - Structured        | <b>0.0363</b> | 0.0647        | <b>0.0299</b> | <b>+4.76%</b>          |
| T4 - User-Centric      | 0.0359        | 0.0633        | 0.0296        | +3.40%                 |
| T5 - Comparative       | 0.0358        | 0.0653        | 0.0291        | +3.32%                 |
| T7 - Struct+Comp.      | 0.0357        | <b>0.0662</b> | 0.0286        | +2.90%                 |

These results confirm that the LLM2Sequential framework can be meaningfully improved without modifying either the embedding model or the downstream recommendation architecture-solely by redesigning the textual description passed to the encoder. The gains are modest but consistent, which is expected given that the

description strategy affects only the first stage of the pipeline and that the BGE-M3 encoder is robust to moderate input variations. This finding positions description strategy selection as a practical, low-cost lever for improving LLM-based sequential recommendation systems in production environments.

## 5.6 Discussion

The experimental results support three main findings.

**Finding 1 - Structured metadata yields the single largest improvement.**

The T2 strategy (Structured Attributes) achieves the best NDCG@10, NDCG@20, and MRR, demonstrating that a simple, mechanical enrichment of item names with category and brand fields provides the strongest and most consistent benefit. This finding has practical implications: for e-commerce platforms where rich metadata is readily available, the structured attribute prompt represents a low-cost, high-return improvement over the default title-only approach. The magnitude of the gain (+4.76% in NDCG@10) is approximately double that of most alternative strategies, underscoring the importance of well-structured, concise metadata over free-form text.

**Finding 2 - Cross-item relational descriptions are beneficial but not transformative.** Both strategies that incorporate comparative references to similar items (T5 and T7) outperform the T1 baseline, ranking among the top four strategies. This suggests that the embedding model benefits from explicit relational context: when an item’s description references other items, the resulting embedding vector implicitly encodes its position relative to those items, creating a more informative neighborhood structure for SASRec’s attention mechanism. However, these strategies do not surpass the simpler T2, indicating that the structured presentation of an item’s own attributes is more impactful than relational cross-item signals.

**Finding 3 - Rich prose descriptions provide no measurable benefit.** The T3 strategy (Rich Prose) performs essentially at parity with the title-only baseline (−0.61% in NDCG@10). This is an important result: it demonstrates that more text does not automatically lead to better embeddings. Natural language descriptions introduce variability in length, style, and information density across items, which may offset any informational benefit. Unlike structured attributes, which present information in a consistent, predictable format, prose descriptions vary across items, producing an embedding space where items are not compared on a uniform basis. The negligible gap suggests that rich prose neither helps nor significantly hurts—it is simply a neutral design choice.

These findings have direct implications for the design of LLM-based recommendation

systems. Practitioners should consider the description strategy as a first-class design choice rather than a trivial implementation detail. Based on our results, we recommend the Structured Attributes strategy (T2) as the default choice for its simplicity, robustness, and top performance, with the Comparative strategy (T5) as a viable alternative when HR@20 is the priority metric.

## 5.7 Conclusion

This chapter evaluated seven item description strategies for the `LLM2Sequential` pipeline on the Amazon Beauty dataset. The results confirm that the choice of description strategy has a measurable impact on downstream recommendation accuracy, with the best strategy (T2 - Structured Attributes) improving NDCG@10 by +4.76% over the title-only baseline of [Boz et al., \(2025\)](#). The Rich Prose strategy (T3) performs at parity with the baseline (−0.61%), demonstrating that simply adding more text does not guarantee improvement and that description *format* matters more than description *length*.

The intrinsic embedding quality analysis revealed that average pairwise cosine distance does not directly predict recommendation performance, suggesting that the downstream model’s ability to exploit embedding structure depends on the alignment between semantic similarity and collaborative patterns rather than on geometric spread alone. The direct comparison with [Boz et al., \(2025\)](#) confirms that description strategy selection is a practical, low-cost improvement lever that requires no architectural changes. Together, these findings establish a first evidence-based guideline for selecting item description strategies in LLM-based sequential recommendation and open several directions for future work, including the extension to additional datasets (Steam, MovieLens) and the exploration of dataset-adaptive prompt strategies that automatically select the best description format based on available metadata.

# Conclusion Générale et Perspectives

## Synthèse des contributions

Ce mémoire a présenté deux contributions complémentaires dans le domaine de la recommandation séquentielle basée sur les modèles de langage de grande taille (LLMs).

La **première contribution** a proposé et évalué une architecture hybride GNN-Transformer à fusion adaptative selon la longueur d'historique (*Length-Adaptive Hybrid GNN-Transformer*). Le modèle construit un graphe global de co-occurrence entre items via un encodeur GCN, dont les représentations sont fusionnées avec celles de BERT4Rec grâce à un poids  $\alpha(u)$  défini par la longueur de l'historique de l'utilisateur. Les expériences menées sur MovieLens-100K et MovieLens-1M ont montré que tous les variants hybrides surpassent ou égalent BERT4Rec, SASRec, GRU4Rec, Caser et LightGCN sur HR@10, NDCG@10 et MRR@10, tout en n'ajoutant que 5.9% de paramètres supplémentaires.

La **deuxième contribution** a mené une étude systématique de sept stratégies de description textuelle des items pour le pipeline LLM2Sequential (Boz et al., 2025), évaluées sur le dataset Amazon Beauty avec SASRec comme modèle de recommandation. La stratégie T2 (Structured Attributes) améliore le NDCG@10 de +4.76% par rapport au baseline titre-seul, tandis que la stratégie T3 (Rich Prose) ne montre aucune amélioration (−0.61%).

## Apports principaux

Premièrement, l'architecture du modèle de fusion doit être **adaptative au niveau utilisateur**. Deuxièmement, la **stratégie de description textuelle des items** est un levier d'amélioration sous-exploité dans les pipelines LLM-based. Troisièmement, les deux contributions montrent qu'il est possible d'améliorer les performances sans modification architecturale majeure.

## Limites

Les travaux présentés comportent plusieurs limites. La première contribution est évaluée exclusivement sur des données MovieLens; la généralisation à des domaines

plus parcimonieux reste à vérifier. Pour la deuxième contribution, l'évaluation est limitée à un seul dataset (Amazon Beauty) et à un seul modèle (SASRec).

## Perspectives

**Axe 1 - Fusion continue et différentiable.** Remplacer les bins discrets de Hybrid Discrete par un MLP entièrement différentiable qui produit  $\alpha(u)$  comme une fonction continue de  $L(u)$ .

**Axe 2 - Généralisation des stratégies de description.** Étendre l'étude des stratégies de description aux datasets Steam et MovieLens.

**Axe 3 - Principe adaptatif au-delà de la recommandation.** Investiguer si le principe de fusion adaptative peut être transposé à d'autres problèmes de fusion multimodale.



## Productions Scientifiques

### Conference Papers

1. **Farouk Ouled Meriem**, Samira Lagrini, Sarra Benzaid, and Nada Zeraoui.  
“Dynamic Sequence Modeling via Hybrid GNN-Transformer Architecture for Robust Sequential Recommendation.” *Proceedings of The 2nd National Conference on Artificial Intelligence and Digital Transition (CAIDT'26)*, Kasdi Merbah University, Ouargla, Algeria, April 29–30, 2026.



Figure 5.1: Certificate of participation for the paper "Dynamic Sequence Modeling via Hybrid GNN-Transformer Architecture for Robust Sequential Recommendation" presented at the 2nd National Conference on Artificial Intelligence and Digital Transition (CAIDT'26), University of Kasdi Merbah, Ouargla, Algeria, April 29–30, 2026.

## General Conclusion and Future Work

This memoir addressed two open problems in sequential recommendation systems: the adaptive fusion of graph-based and sequence-based representations for users with heterogeneous interaction histories, and the systematic evaluation of item textual description strategies for LLM embedding generation. The work was structured around two main contributions, each validated through empirical experiments on standard benchmark datasets.

**Contribution 1: Length-Adaptive Hybrid GNN-Transformer.** We proposed a family of hybrid models that combine a GNN encoder (LightGCN) with a Transformer encoder (BERT4Rec) through a user-specific fusion mechanism based on interaction history length. The core insight is that users with sparse histories benefit more from global graph-based collaborative signals, while users with rich histories can better exploit detailed sequential patterns. This insight was formalized through a fusion weight  $\lambda(u)$  that varies according to the length of each user’s interaction history, and was implemented in two variants: HybridFixed with a learnable global fusion weight, and HybridDiscrete with length-dependent bin thresholds.

The experimental evaluation on MovieLens-100K and MovieLens-1M demonstrated that the proposed hybrid variants consistently outperform strong sequential and graph-based baselines. HybridFixed achieved improvements of +8.1% in HR@10, +11.5% in NDCG@10, and +14.2% in MRR over the BERT4Rec baseline on ML-1M. The per-bucket analysis confirmed that the adaptive fusion mechanism provides targeted gains for short-history users, who are systematically underserved by Transformer-only models. The ablation study further confirmed that both the GNN branch and the Transformer branch contribute independently to performance, and that the fusion strategy itself is a critical design choice.

**Contribution 2: PromptCraft-SeqRec - Systematic Evaluation of Item Description Strategies.** We designed and evaluated seven item description strategies for the LLM2Sequential pipeline, spanning from minimal title-only encoding to rich hybrid descriptions incorporating structured attributes, user-centric framing, and comparative references. Each strategy embodies a distinct hypothesis about the information requirements of LLM-based sequential recommendation.

The experiments on the Amazon Beauty dataset revealed that the choice of description strategy measurably affects downstream recommendation accuracy. The

Structured Attributes strategy (T2) achieved the best overall NDCG@10 with a +4.76% improvement over the title-only baseline, while also delivering the best MRR. The Comparative (T5) and User-Centric (T4) strategies also provided meaningful gains. Critically, the Rich Prose strategy (T3) performed at parity with the baseline (−0.61%), establishing that providing more text does not necessarily yield better embeddings and that the *format* of item descriptions is as important as their *content*. These findings provide the first evidence-based guideline for selecting item description strategies in LLM-based recommendation systems.

**Limitations.** For Contribution 1, the length-adaptive fusion mechanism relies on hand-crafted thresholds and discrete binning, which may not capture every user’s optimal balance between graph and sequence signals. For Contribution 2, the experiments were conducted on a single dataset (Amazon Beauty) and a single recommendation model (SASRec). The generalizability of the findings to other domains with different metadata profiles remains an open question.

**Future Directions.** First, extending the adaptive fusion mechanism to a fully learnable, continuous function could further improve performance across heterogeneous user populations. Second, evaluating the PromptCraft strategies on additional datasets (Steam, MovieLens, Delivery Hero) would establish the extent to which findings generalize across domains. Third, exploring dataset-adaptive prompt selection could make the PromptCraft framework practical for real-world deployment without manual strategy tuning. Fourth, investigating the interaction between prompt strategy and the choice of embedding model could reveal whether certain strategies benefit more from larger or differently-trained embedding models. Finally, extending the PromptCraft methodology to other recommendation paradigms beyond sequential recommendation would broaden the impact of the finding that item description design is a first-class choice in LLM-augmented recommendation systems.

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